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**TITLE:** GXP and Non-GXP Study Report Review and Approval Form

| DOCUMENT NO. | VERSION | ISSUE DATE | EFFECTIVE DATE | PAGE |
|---|---|---|---|---|
| GE003FM01 | 02 | 17 Oct 2014 | **20 Oct 2014** | 2 of 2 |

☐ Vendor Study Protocol ☑ Vendor Study Report ☐ Lpath Study Report

| | |
|---|---|
| Lpath Study No.: LT3114-PHA-018-R | Version No.: 01 ☑ Non GXP ☐ GLP ☐ GCP ☐ GMP |
| Product Name: LT3114 | Test Article: B3 (504B3); LT1015 |

Study Title: Efficacy of IV 504B3 on primary hyperalgesia in the Brennan model of post-incisional pain in rats (non-GLP)

| Study Vendor/Site: ☐ N/A | Study Director: ☐ N/A | Study Contributor(s): ☐ N/A |
|---|---|---|
| Calvert, Scott Township PA | Rosalia Matteo | Ann Gregus; Stephanie Knapik |

| Vendor Study No./Version: ☐ N/A | Study Dates: | Report Date: |
|---|---|---|
| 1274RL35.001 | Feb 2014 - May 2014 | Nov 2014 |

Lab Notebook Reference: ☑ N/A

Executive summary of study results: ☐ See Attached ☐ N/A

The efficacy of the anti-LPA antibody, B3, was examined in a post-surgical incision model (Brennan). Sprague Dawley rats were anesthetized and an incision was made in the plantar aspect of the right paw and the incision closed. Treatments were administered 1h or 3 h post surgery and on days 3 and 6 for B3 and the antibody control LT1015. Paw withdrawal measurements were carried out one day prior to surgery, on the day of surgery and 1,3, 6 an d9 days post surgery.
Intravenous administration of LT1015 or 504B3 at 10 or 30mg/kg, starting 1h before or 3 h post surgery) did not result in any statistical increase in paw withdrawal threshold in either the left or right paw. However the positive control agent (morphine sulfate, 3mg/kg sc) produced significant increases in the mean withdrawal response in both paws.

**LPATH APPROVAL**

| Function | Name (Print) | Signature | Date |
|---|---|---|---|
| Study Director/Monitor | Rosalia Matteo | [signature] | 11/18/2014 |
| Contributor | Ann M. Gregus<br>Stephanie Knapik | [signature]<br>[signature] | 11/18/2014<br>11/18/2014 |
| Research | GARY WOODNUTT | [signature] | 11/18/2014 |
| Development | DARIO PAGGIARINO | [signature] | 11/19/2014 |
| Quality Assurance | ELVIRA MARQUEZ | [signature] | 26NOV14 |

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# Final Report

| | |
|---|---|
| **Title:** | Efficacy of IV 504B3 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP) |
| **Calvert Study No.:** | 1274RL35.001 |
| **Testing Facility:** | Calvert Laboratories, Inc.<br>Scott Technology Park<br>130 Discovery Drive<br>Scott Township, PA 18447 |
| **Study Sponsor:** | Lpath, Inc.<br>4025 Sorrento Valley Blvd.<br>San Diego, CA 92121-1404 |
| **Date:** | 14 Nov 2014 |

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# *I. Table of Contents*

# II. Signatures

Calvert Study No.: 1274RL35.001

Title: Efficacy of IV 504B3 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

## A. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained.

Brian Klatt ______ 14 NOV 2014

Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

Date

## B. Signatures of Other Responsible Personnel

Scientific Oversight and Report Review

______ 14 Nov 2014

Scientific Management
Calvert Laboratories, Inc.

Date

# *III. Summary*

## A. Title

Efficacy of IV 504B3 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

## B. Objective

The purpose of the study is to evaluate the potential ability of a test article to produce an analgesic response to post-surgical pain in rats.

## C. Methods

The study was conducted with 8 groups of ten male Sprague Dawley rats each. The animals were anesthetized with isoflurane (4%) and a 1 cm incision was made in the plantar aspect of the rat's right hind paw. The flexor muscle was elevated and incised longitudinally via blunt dissection with the muscle origin and insertion remaining intact. The incision was closed with two sutures and the rats were allowed to recover in their cages. The test article, reference article and vehicle were administered three times to each animal according to the following table:

| Group (n=10) | Surgery | Oral Treatment | Blinded Treatment | Route | Dose (mg/kg) | Volume (mL/kg) | Dose Schedule |
|---|---|---|---|---|---|---|---|
| 2 | Sham | Vehicle control (PBS) | B | IV | 0 | 2.40 | +3h, d3, d6 |
| 1 | Brennan | LT1015 | A | IV | 10 | 2.35 | -1h, d3, d6 |
| 4 | Brennan | 504B3 | D | IV | 10 | 2.40 | -1h, d3, d6 |
| 5 | Brennan | LT1015 | E | IV | 10 | 2.35 | +3h, d3, d6 |
| 7 | Brennan | 504B3 | G | IV | 10 | 2.40 | +3h, d3, d6 |
| 3 | Brennan | LT1015 | C | IV | 30 | 2.35 | +3h, d3, d6 |
| 6 | Brennan | 504B3 | F | IV | 30 | 2.40 | +3h, d3, d6 |
| 8 | Brennan | Morphine sulfate | - | SC | 3.0 | 1 | +3h, d3, d6 |

Treatments were administered 1 hour prior to or three hours after surgery on Day 0 and on Days 3 and 6. Dosing on Days 3 and 6 were after the behavioral testing. Paw withdrawal thresholds were determined using von Frey filaments one day prior to surgery, on the day of surgery and on Days 1, 3, 6 and 9 after surgery. Paw withdrawal thresholds were conducted on both hind paws by an observer blinded to the treatment conditions. Whole blood was collected by cardiocentesis immediately following the behavioral testing on Day 9. The Sponsor will provide contact information for shipping the samples.

## D. Results

The intravenous administration of LT1015 or 504B3 at 10 or 30 mg/kg did not produce any statistically significant increases in mean withdrawal response in either the left or right hindpaw on the day of surgery or on Days 1, 3 and 6 when compared to the morphine predose values on the respective days. There were no differences observed whether the first dose of LT1015 or 504B3 at 10 mg/kg was administered prior to surgery or after surgery.

Morphine sulfate administered at 3 mg/kg by subcutaneous injection produced increases in mean withdrawal response in both left and right hind paws at 30, 60, 90 and 120 minutes following dosing on Days 0, 3 and 6 when compared to the predose values on the respective days. The increases in mean withdrawal response were statistically significant ($p \leq 0.05$) compared to the predose values with the exception of the left hind paw at 120 minutes post-dose on Day 0, the left hind paw at 90 and 120 minutes post-dose on Day 3, and the right hind paw at 120 minutes post-dose on Day 3.

## E. Conclusion

Based on the results of this study, the intravenous administration of LT1015 and 504B3 at 10 and 30 mg/kg did not produce an analgesic response to post-surgical pain in rats. Morphine sulfate administered at 3 mg/kg by subcutaneous injection produced significant increases in the mean withdrawal response.

# IV. General Information

## A. Key Study Dates

| | |
|---|---|
| Study Initiation: | 21 Feb 2014 |
| Animal Receipt Dates: | 7 and 14 May 2014 |
| First Day of Dosing: | 13 May 2014 |
| Last Day of In-Life: | 29 May 2014 |

## B. Responsible Personnel at Testing Facility

| | |
|---|---|
| Study Director: | Brian Klatt, B.S. |
| Primary Technicians: | Brian Klatt, B.S.<br>Jennifer Shenise, A.S., ALAT<br>Danielle Pallman, B.S., LVT<br>Laurie M. Serfilippi, VMD, DACLAM, CPIA<br>Jennifer Shenise, A.S., ALAT |

## C. Contributing Scientist

| | |
|---|---|
| Study Monitor: | Rosalia Matteo, Ph.D.<br>Lpath, Inc.<br>4025 Sorrento Valley Blvd.<br>San Diego, CA 92121-1404<br>Phone: (858) 678-0800 ext 112<br>Fax: (858) 678-0900<br>Email: RMatteo@lpath.com |

## D. Objective

The purpose of the study is to evaluate the potential ability of a test article to produce an analgesic response to post-surgical pain in rats.

## E. Experimental Design Overview

### *1. General Description*

The study was conducted with 8 groups of ten male Sprague Dawley rats each. The animals were anesthetized with isoflurane (4%) and a 1 cm incision was made in the plantar aspect of the rat's right hind paw. The flexor muscle was elevated and incised longitudinally via blunt dissection with the muscle origin and insertion remaining intact. The incision was closed with two sutures and the rats were allowed to recover in their cages. The test article, reference article and vehicle were administered three times to each animal according to the following table:

| Group (n=10) | Surgery | Oral Treatment | Route | Dose (mg/kg) | Volume (mL/kg) | Dose Schedule |
|---|---|---|---|---|---|---|
| 2 | Sham | Vehicle control (PBS) | IV | 0 | 2.40 | +3h, d3, d6 |
| 1 | Brennan | LT1015 | IV | 10 | 2.35 | -1h, d3, d6 |
| 4 | Brennan | 504B3 | IV | 10 | 2.40 | -1h, d3, d6 |
| 5 | Brennan | LT1015 | IV | 10 | 2.35 | +3h, d3, d6 |
| 7 | Brennan | 504B3 | IV | 10 | 2.40 | +3h, d3, d6 |
| 3 | Brennan | LT1015 | IV | 30 | 2.35 | +3h, d3, d6 |
| 6 | Brennan | 504B3 | IV | 30 | 2.40 | +3h, d3, d6 |
| 8 | Brennan | Morphine sulfate | SC | 3.0 | 1 | +3h, d3, d6 |

Treatments were administered 1 hour prior to or three hours after surgery on Day 0 and on Days 3 and 6. Dosing on Days 3 and 6 were after the behavioral testing. Paw withdrawal thresholds were determined using von Frey filaments one day prior to surgery, on the day of surgery and on Days 1, 3, 6 and 9 after surgery. Paw withdrawal thresholds were conducted on both hind paws by an observer blinded to the treatment conditions. Whole blood was collected by cardiocentesis immediately following the behavioral testing on Day 9. The Sponsor will provide contact information for shipping the samples.

# *V. Materials and Methods*

## A. Test Article, Reference Article and Vehicles

### *1. Test Article 1*

Identification: LT1015 (12.74 mg/mL)

Lot/Batch No.: HR121332

Physical Description: Clear, colorless liquid

Storage Conditions: Stock vials were stored at -80°C. Fresh dilutions were prepared prior to dosing in PBS as needed and stored at 2-8°C for up to 1 week. Do not freeze/thaw.

### 2. *Test Article 2*

Identification: 504B3 (12.48 mg/mL)

Lot/Batch No.: HR121338

Physical Description: Clear, colorless liquid

Storage Conditions: Stock vials were stored at -80°C. Fresh dilutions were prepared prior to dosing in PBS as needed and stored at 2-8°C for up to 1 week. Do not freeze/thaw.

### 3. *Reference Article*

Identification: Morphine sulfate (15 mg/mL)

Source/Manufacturer: Westward

Lot/Batch No.: 043323 (Exp. date: Apr 2015)

Physical Description: Clear colorless liquid

Storage Conditions: Room temperature

Preparation: A 3 mg/mL formulation was prepared on 13, 16, 19, 20, 23 and 26 May 2014 as follows:

Using aseptic techniques in a cleaned laminar flow hood, the required amount of morphine sulfate was measured using an appropriately-sized sterile syringe and sterile 18 gauge needles and dispensed into an appropriately-sized amber glass vial. The required amount of saline was measured using an appropriately-sized sterile syringe and added to the vial. The vial was sealed with a sterile rubber stopper and crimp cap and repeatedly inverted until a homogeneous formulation was achieved. Once homogeneous, the formulation was d stored at room temperature until needed for dosing. On 13 May 2014, the pH (6) was taken using Litmus paper.

### *4. Vehicle*

| | |
|---|---|
| Identification: | Phosphate buffered saline (PBS) |
| Source/Manufacturer: | Sigma |
| Lot/Batch No.: | SLBD4322V (Exp. date: Feb 2016) |
| Physical Description: | White tablets |
| Storage Conditions: | Room temperature |
| Preparation: | One PBS tablet was placed in an appropriately sized and labeled amber glass bottle. 200 mL of sterile water for injection was measured using a graduated cylinder and added to the bottle containing the tablet. A stir bar was added and the bottle was placed on a magnetic stir plate to mix until dissolution. Once dissolved, the solution was sterile filtered using a 0.22 µm PVDF syringe filter into a sterile glass vial and stored at room temperature until use. The clear, colorless liquid was prepared on 12 May 2014 and was assigned an expiration date of 26 May 2014. On the first day of preparation, the pH (7) was taken using Litmus paper. On 12 and 19 May 2014, 10.5 mL of PBS was aseptically divided among three sterile amber glass vials (approximately 3.5 mL each), sealed with sterile rubber stoppers and crimp caps, and stored refrigerated at 2-8°C until needed for dosing. |

### *5. Dose Preparation*

The test article formulations were provided as liquids by the Sponsor and dosed as received for the 30 mg/kg dose groups. For the 10 mg/kg dose groups, the stocks solutions were diluted to concentrations that allowed the same dose volumes as the 30 mg/kg dose groups as follows:

| Date | Test Article | Desired Concentration (mg/mL) | Amount of 12.74 mg/mL stock added (mL) | Amount of PBS added (mL) |
|---|---|---|---|---|
| 12 May 2014 | LT1015 | 4.26 | 7.02 | 13.98 |
| 19 May 2014 | LT1015 | 4.26 | 7.02 | 13.98 |

| Date | Test Article | Desired Concentration (mg/mL) | Amount of 12.48 mg/mL stock added (mL) | Amount of PBS added (mL) |
|---|---|---|---|---|
| 12 May 2014 | 504B3 | 4.17 | 7.02 | 13.98 |
| 19 May 2014 | 504B3 | 4.17 | 7.02 | 13.98 |

Using aseptic techniques in a cleaned laminar flow hood, the required amount of test article was measured using an appropriately-sized sterile syringes and sterile 18 gauge needles and dispensed into an appropriately-sized amber glass bottle. The required amount of PBS was measured using an appropriately-sized sterile syringe and added to the bottle. The vial was sealed with a sterile rubber stopper and crimp cap and repeatedly inverted until a homogeneous formulation was achieved. Once homogeneous, the formulation was divided among six sterile amber vials (approximately 3.5 mL each) and stored refrigerated at 2-8°C until needed for dosing. On 12 and 19 May 2014, 10 vials of the test article stock solutions (12.74 or 12.48 mg/mL) were aliquotted into 3 appropriately labeled sterile amber vials (approximately 3.5 mL each) and stored refrigerated at 2-8°C until needed for dosing. PBS was aseptically divided among three sterile amber glass vials (approximately 3.5 mL each), sealed with sterile rubber stoppers and crimp caps, and stored refrigerated at 2-8°C until needed for dosing.

On 12 May 2014, the pH of each formulation was taken using Litmus paper. All test article formulations had a pH of 7.

Upon completion, all formulations were labeled “Treatment A” through Treatment G” for the purpose of “blinding” the observer to the treatment groups. Decoding of the treatment groups occurred after the final in-life observations.

Formulations were aliquotted for the week and kept at room temperature until needed for dosing. Unused portions were properly discarded after dosing.

## B. Test System (Animals and Animal Care)

### *1. Description*

| | |
|---|---|
| Species: | Rat |
| Stock: | Hsd: SD |
| Total Number: | Eighty (80): ten (10) per group |
| Gender: | Male |
| Age Range: | 7-8 weeks |
| Body Weight Range: | 192-229 grams |
| Animal Source: | Harlan |
| Experimental History: | Purpose-bred and experimentally naïve at the outset of the study |
| Identification: | Ear tag number and color code markings |

### *2. Rationale for Choice of Species and Number of Animals*

The rat is a standard species used for the evaluation of potential analgesic properties of a test article (1-6).

Based on Calvert's experience with this model and available literature, the number of animals used in this study is the minimum amount considered necessary to provide valid scientific assessment (4,5,6).

### *3. Husbandry*

| | |
|---|---|
| Housing: | Animals were group housed upon receipt in compliance with the National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals were kept was documented in the study records. No other species were kept in the same room. The experiment was |

conducted in a room dedicated to this study and the dosing occurred in this room only. Following surgery, animals were housed in totes with bedding.

| | |
|---|---|
| Lighting: | 12 hour light/12 hour dark |
| Room Temperature: | 22 to 23°C |
| Relative Humidity: | 42 to 58% |
| Food: | All animals had access to Certified Rodent Diet (TEKLAD) *ad libitum*. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants were known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which are maintained in the Calvert archives. |
| Water: | Water was available *ad libitum,* to each animal, via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants were known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives. |
| Acclimation: | Study animals were acclimated to their housing for six days prior to their first day of dosing. |

### 4. *Prestudy Health Screen and Selection Criteria*

All animals received for this study were assessed as to their general health by a member of the veterinary staff or other suitably trained authorized personnel. During the acclimation period, each rat was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use were assigned to this study.

### *5. Assignment to Study Groups*

Rats were selected by body weight and apparent good health and randomly assigned to study groups. Animals will be distributed evenly into groups according to their average baseline von Frey thresholds as follows:

| Group Number | Treatment | Males |
|---|---|---|
| 1 | A | 1701-1710 |
| 2 | B | 1711-1720 |
| 3 | C | 1721-1730 |
| 4 | D | 1731-1740 |
| 5 | E | 1741-1750 |
| 6 | F | 1751-1760 |
| 7 | G | 1761-1770 |
| 8 | Morphine | 1771-1780 |

### *6. Humane Care of Animals*

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 2011, National Academy Press). The Calvert IACUC approved the study protocol prior to finalization.

## C. Test Article Administration

### *1. Group Assignments and Dose Levels*

| Group | Blinded Treatment | Surgery | Test Article | Dose (mg/kg in 1 ml/kg) | Dose Volume (mL/kg) | Route | Dose Schedule | Number of rats |
|---|---|---|---|---|---|---|---|---|
| 2 | B | Sham | PBS | 0 | 2.40 | IV | +3h, d3, d6 | 10 |
| 1 | A | Brennan | LT1015 | 10 | 2.35 | IV | -1h, d3, d6 | 10 |
| 4 | D | Brennan | 504B3 | 10 | 2.40 | IV | -1h, d3, d6 | 10 |
| 5 | E | Brennan | LT1015 | 10 | 2.35 | IV | +3h, d3, d6 | 10 |
| 7 | G | Brennan | 504B3 | 10 | 2.40 | IV | +3h, d3, d6 | 10 |
| 3 | C | Brennan | LT1015 | 30 | 2.35 | IV | +3h, d3, d6 | 10 |
| 6 | F | Brennan | 504B3 | 30 | 2.40 | IV | +3h, d3, d6 | 10 |
| 8 | - | Brennan | Morphine | 3 | 1.0 | SC | +3h, d3, d6 | 10 |

### *2. Dosing*

Route: Intravenous (i.v.) injection for the vehicle and test articles. The injection was administered as a slow bolus over approximately 30 seconds.

Subcutaneous (s.c.) injection for the reference article

Frequency: Three times. 1 hour prior to or 3 hours after surgery, then twice a week (on Days 3 and 6) after behavioral testing.

Procedure: Vehicle or test articles were administered by intravenous injection at 2.35 or 2.40 mL/kg. Reference article was administered by subcutaneous injection at 1 mL/kg.

### 3. *Justification for Route and Dose Levels*

The route of administration was chosen as optimal based on pharmacokinetic studies of blood levels and half-life.

Doses were selected based on pharmacological efficacy studies in a variety of rat models of disease.

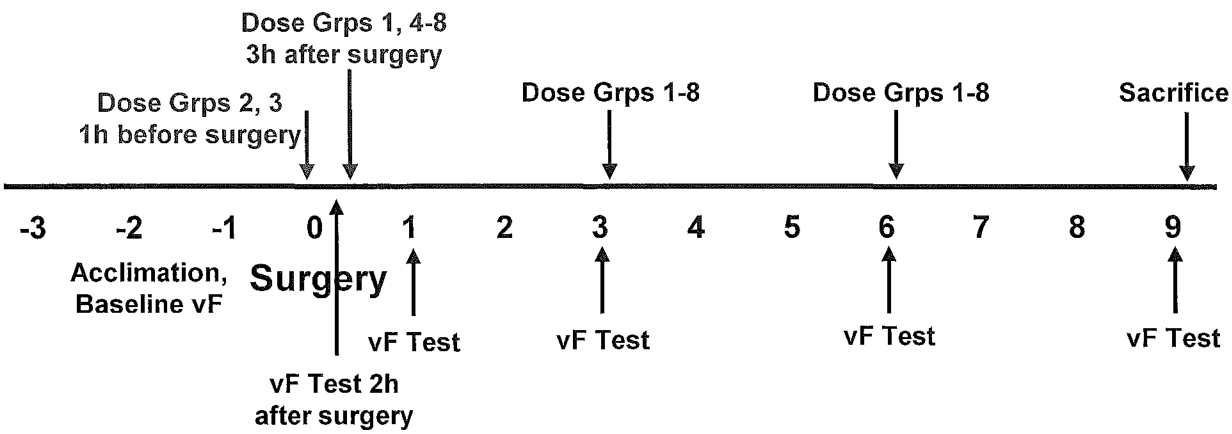
## D. Method of Study Performance

## A. Method of Study Performance

### 1. *Study Details:*

| Procedure | Time Points | Notes |
|---|---|---|
| Acclimation | 3-5 days before surgery | |
| von Frey behavioral testing | 1-3 days before surgery, then 2h, and on days 1, 3, 6, 9 after surgery | Also 30 min-2h after dosing morphine (see description) |
| Body weights | Prior to surgery, then every 2-3 days thereafter | |
| Antibody dosing | 1h prior to or 3h after surgery, then twice/week (Days 3 and 6) after behavioral testing | |
| Brennan or sham surgery | Day 0 | |
| Clinical observations | Daily for morbidity/mortality | |
| Report | Standard in-life summary | |
| Terminal Procedures | End of study | Collect blood for LPA measurement |
| Archive | | 2 years |

**Timeline (days):**

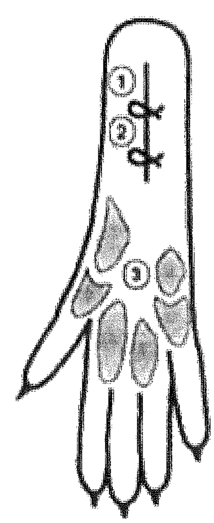



### 2. *Behavioral Testing:*

The rats were placed individually on an elevated wire mesh floor covered with a clear plastic cage top and allowed to acclimate for ≥30 min to the von Frey testing apparatus and baseline tactile thresholds were measured. Measurements of tactile paw withdrawal thresholds (PWT) were conducted on both hindpaws by an observer blinded to the treatment conditions. Paw withdrawal responses were determined using von Frey filaments applied from underneath the cage through openings. Each von Frey filament was applied at the medial side of the wound near the heel (site 1, see diagram below) once starting with a 1 or 2 g filament and continuing until a withdrawal response occurred or 60 g force was reached. The median force producing a response, determined from three tests given over a 10-min period, was considered the withdrawal threshold. Animals were distributed evenly into groups according to their average baseline von Frey thresholds. Testing for morphine occurred at 30, 60, 90 and 120 minutes after injection.

### 3. *Surgical Narrative*

Animals were weighed prior to surgery and every 2-3 days thereafter. Surgical procedures were performed under deep isoflurane anesthesia (3-4% at 3-4 L/min $O_2$). All surgical procedures were performed at the surgical planes of anesthesia, using sterile instruments that are re-sterilized for each animal. After disinfection of the right hind paw with Betadine (povidone-iodine) and 70% isopropyl alcohol, rats received an intramuscular injection of penicillin (30,000 IU) in the triceps muscle. A 1 cm longitudinal incision was made with a number 11 blade through the



skin and fascia of the right hind paw starting 0.5 cm from the proximal edge of the heel and extending toward the toes (1). Following skin incision, blunt curved forceps were used to raise and retract the plantar flexor digitorum brevis muscle. The plantaris muscle was then incised longitudinally, keeping the muscle origin and insertion intact. The skin was closed with 2 mattress sutures of 5-0 silk, and the wound site was covered with triple antibiotic ointment (polymixin B, neomycin and bacitracin). To minimize distress, fluids (s.c. Plasmalyte – 1 ml/50g) was administered immediately post-operatively. Sham surgeries consisted of all procedures except incision and muscle retraction (anesthesia, antibiotics and surgical preparation). After surgery, the animals were allowed to recover in a box with a heating pad so as to prevent rapid loss of body heat following surgery. On the second postoperative day, sutures were removed according to Brennan et al. Animals were monitored daily for general health. No animals showed any evidence of motor dysfunction, > 20% weight loss or wound dehiscence after surgery.

### 4. *Blood Collection*

Immediately following the behavioral testing on Day 9, the rats were anesthetized by $CO_2$ inhalation and whole blood samples (approximately 0.5 ml/sample) were collected by cardiocentesis into blood collection tubes containing EDTA. Immediately following blood collection, the animals were euthanized by $CO_2$ asphyxiation. Blood samples were spun in a centrifuge (3000 rpm for approximately 10 minutes). After centrifugation, plasma was collected and stored at -80°C ± 10°C until shipped for analysis. The plasma samples were shipped on dry ice to:

Rosalia Matteo, Ph.D.
Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

The Mouse IgG ELISA methodology and results of the plasma analysis are included in Appendix II.

## B. Terminal Procedures

### *1. Termination*

a) <u>Scheduled Sacrifice</u>

All animals were euthanized by $CO_2$ asphyxiation immediately following the blood collection on Day 9.

# *VI. Records and Reports*

## A. Statistical Analysis

Mean and standard error of the mean will be calculated for paw withdrawal thresholds. Statistically significant effects will be determined using an ANOVA with Tukey HSD Multiple Comparison Test (Systat V. 9.01). A p value $\leq 0.05$ will denote statistical significance.

## B. Storage of Records

In-life data, protocol, protocol amendments (if applicable), and the final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

# *VII. Results*

As presented in Table1, the intravenous administration of LT1015 or 504B3 at 10 or 30 mg/kg did not produce any statistically significant increases in mean withdrawal response in either the left or right hindpaw on the day of surgery or on Days 1, 3 and 6 when compared to the morphine predose values on the respective days. There were no differences observed whether the first dose of LT1015 or 504B3 at 10 mg/kg was administered prior to surgery or after surgery.

Morphine sulfate administered at 3 mg/kg by subcutaneous injection produced increases in mean withdrawal response in both left and right hind paws at 30, 60, 90 and 120 minutes following dosing on Days 0, 3 and 6 when compared to the predose values on the respective days. The increases in mean withdrawal response were statistically significant ($p \leq 0.05$) compared to the predose values with the exception of the left hind paw at 120 minutes post-dose on Day 0, the left hind paw at 90 and 120 minutes post-dose on Day 3, and the right hind paw at 120 minutes post-dose on Day 3.

Individual animal data is presented in Appendix I.

## *VIII. Conclusion*

Based on the results of this study, the intravenous administration of LT1015 and 504B3 at 10 and 30 mg/kg did not produce an analgesic response to post-surgical pain in rats. Morphine sulfate administered at 3 mg/kg by subcutaneous injection produced significant increases in the mean withdrawal response.

## *IX. References*


1. Brennan, T.J., Vandermeulen, E.P. and Gebhart, G.F. Characterization of a rat model of incisional pain. Pain **64**: 493-501 (1996).

2. Hamalainen, M.M., Gebhart, G.F. and Brennan, T.J. Acute effects of an incision on mechanosensitive afferents in the plantar rat hindpaw. *J Neurophysiol* **87**: 712-720(2002).

3. Zahn, P.K. and Brennan, T.J. Primary and secondary hyperalgesia in a rat model for human post-operative pain. Anesthesiology **90**: 863-872(1999).

4. Whiteside, G.T., Harrison, J., Boulet, J., Mark, L., Pearson, M., Gottshall, S. and Walker, K., Pharmacological characterization of a rat model of incisional pain. British Journal of Pharmacology **141**: 85-91(2004).

5. Xu, J. and Brennan, T.J., Guarding Pain and Spontaneous Activity of Nociceptors after Skin versus Skin Plus Deep Tissue Incision, Anesthesiology 112:153-164 (2010).

6. Brennan, T.J., Postoperative Models of Nociception, ILAR Journal **40(3)**:129-136 (1999).

# X. Tables

## A. Table 1 - Mean Withdrawal Response to von Frey Filaments

| Group (n=10) | Treatment[a] | Dose (mg/kg) | Volume (mL/kg) | Route of Administration | Mean Withdrawal Response (g) | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Pre-Surgery | | Day 0 (2 hr post-surgery) | |
| | | | | | Left Paw | Right Paw | Left Paw | Right Paw |
| 2 | Sham (0.5% MC) | 0 | 0 | IV | 15.8 ± 0.66 | 14.5 ± 0.79 | 16.2 ± 2.02 | 14.5* ± 1.79 |
| 1 | LT1015 | 10 | 10 | IV | 13.5 ± 0.66 | 14.3 ± 0.67 | 16.4 ± 2.38 | 3.3 ± 0.44 |
| 4 | 504B3 | 10 | 10 | IV | 14.2 ± 1.62 | 15.5 ± 1.47 | 13.0 ± 1.42 | 3.2 ± 0.36 |
| 5 | LT1015 | 10 | 10 | IV | 14.5 ± 1.55 | 15.5 ± 1.09 | 16.4 ± 1.38 | 3.1 ± 0.39 |
| 7 | 504B3 | 10 | 10 | IV | 14.6 ± 1.45 | 13.4 ± 0.82 | 12.0 ± 0.57 | 3.3 ± 0.61 |
| 3 | LT1015 | 30 | 10 | IV | 13.4 ± 0.89 | 12.2 ± 0.47 | 12.9 ± 1.40 | 2.6 ± 0.38 |
| 6 | 504B3 | 30 | 10 | IV | 14.4 ± 1.56 | 14.1 ± 1.34 | 16.7 ± 1.59 | 3.4 ± 0.28 |
| 8 | Morphine | 3 | 1 | SC | 17.0 ± 1.41 | 16.0 ± 1.63 | 13.2 ± 1.37 | 3.4 ± 0.33 |

Data presented as mean ± standard error of the mean
[a] Doses were administered on the day of surgery (1 hour before surgery for Groups 1 and 4 and 3 hours after surgery for Groups 2, 3 and 5-8) and on Days 3 and 6 after the behavioral testing.
* Statistically significant ($p \le 0.05$) change compared to the respective Group 8 values - ANOVA followed by Tukey HSD Multiple Comparison Test. Group 8 values were obtained prior to morphine dosing on Days 0, 3 and 6..

**A. Table 1 (continued) - Mean Withdrawal Response to von Frey Filaments**

| Group (n=10) | Treatment[a] | Dose (mg/kg) | Volume (mL/kg) | Route of Administration | Mean Withdrawal Response (g) | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Day 1 | | Day 3 | |
| | | | | | Left Paw | Right Paw | Left Paw | Right Paw |
| 2 | Sham (0.5% MC) | 0 | 0 | IV | 14.0 ± 1.11 | 13.4* ± 1.02 | 11.5 ± 1.27 | 13.3* ± 1.65 |
| 1 | LT1015 | 10 | 10 | IV | 13.4 ± 0.78 | 5.2 ± 0.91 | 12.2 ± 0.89 | 4.5 ± 0.50 |
| 4 | 504B3 | 10 | 10 | IV | 9.8 ± 0.82 | 3.5 ± 0.43 | 10.9 ± 0.94 | 5.0 ± 0.61 |
| 5 | LT1015 | 10 | 10 | IV | 12.4 ± 1.22 | 4.2 ± 0.62 | 12.8 ± 1.19 | 5.9 ± 0.63 |
| 7 | 504B3 | 10 | 10 | IV | 11.1 ± 0.57 | 3.9 ± 0.35 | 11.1 ± 1.28 | 4.6 ± 0.47 |
| 3 | LT1015 | 30 | 10 | IV | 13.3 ± 1.33 | 2.9 ± 0.52 | 11.4 ± 0.62 | 6.1 ± 0.65 |
| 6 | 504B3 | 30 | 10 | IV | 12.3 ± 1.61 | 3.7 ± 0.30 | 12.4 ± 0.93 | 6.1 ± 0.66 |
| 8 | Morphine | 3 | 1 | SC | 11.6 ± 1.20 | 3.1 ± 0.32 | 9.7 ± 0.75 | 3.6 ± 0.43 |

Data presented as mean ± standard error of the mean

[a] Doses were administered on the day of surgery (1 hour before surgery for Groups 1 and 4 and 3 hours after surgery for Groups 2, 3 and 5-8) and on Days 3 and 6 after the behavioral testing.

* Statistically significant (p≤0.05) change compared to the respective Group 8 values - ANOVA followed by Tukey HSD Multiple Comparison Test. Group 8 values were obtained prior to morphine dosing. Morphine was dosed on Days 0, 3 and 6.

A. Table 1 (continued) - Summary Data - Mean Withdrawal Response to von Frey Filaments

| Group (n=10) | Treatment[a] | Dose (mg/kg) | Volume (mL/kg) | Route of Administration | Mean Withdrawal Response (g) | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Day 6 | | Day 9 | |
| | | | | | Left Paw | Right Paw | Left Paw | Right Paw |
| 2 | Sham (0.5% MC) | 0 | 0 | IV | 11.1 ± 1.20 | 10.5* ± 0.99 | 9.8 ± 0.46 | 10.6* ± 1.11 |
| 1 | LT1015 | 10 | 10 | IV | 9.5 ± 1.08 | 5.9 ± 0.58 | 11.0 ± 0.92 | 3.9 ± 0.29 |
| 4 | 504B3 | 10 | 10 | IV | 9.4 ± 1.29 | 6.0 ± 0.79 | 8.2 ± 0.48 | 3.6 ± 0.39 |
| 5 | LT1015 | 10 | 10 | IV | 10.4 ± 0.97 | 5.8 ± 0.50 | 9.5 ± 0.76 | 4.6 ± 0.30 |
| 7 | 504B3 | 10 | 10 | IV | 9.9 ± 0.81 | 4.8 ± 0.58 | 10.7 ± 0.94 | 3.9 ± 0.14 |
| 3 | LT1015 | 30 | 10 | IV | 11.5 ± 0.84 | 6.1 ± 0.53 | 9.4 ± 0.82 | 4.5 ± 0.65 |
| 6 | 504B3 | 30 | 10 | IV | 11.3 ± 1.49 | 4.9 ± 0.42 | 9.9 ± 0.81 | 5.1 ± 0.63 |
| 8 | Morphine | 3 | 1 | SC | 8.7 ± 1.01 | 3.5 ± 0.33 | 9.1 ± 0.94 | 2.9 ± 0.33 |

Data presented as mean ± standard error of the mean
[a] Doses were administered on the day of surgery (1 hour before surgery for Groups 1 and 4 and 3 hours after surgery for Groups 2, 3 and 5-8) and on Days 3 and 6 after the behavioral testing.
* Statistically significant ($p \leq 0.05$) change compared to the respective Group 8 values - ANOVA followed by Tukey HSD Multiple Comparison Test. Group 8 values were obtained prior to morphine dosing. Morphine was dosed on Days 0, 3 and 6.

## B. Table 2 - Summary Data - Mean Withdrawal Response to von Frey Filament

| Group (n=10) | Subcutaneous Treatment | Mean Withdrawal Response (g) - Day 0 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Predose | | 30 min pd | | 60 min pd | | 90 min pd | | 120 min pd | |
| | | Left | Right | Left | Right | Left | Right | Left | Right | Left | Right |
| 8 | Morphine sulfate 3 mg/kg 1 mL./kg | 13.2 ± 1.37 | 3.4 ± 0.33 | 50.2* ± 3.65 | 33.2* ± 4.91 | 46.8* ± 4.51 | 32.4* ± 5.89 | 30.4* ± 3.65 | 17.8* ± 1.46 | 19.0 ± 2.14 | 6.4* ± 0.63 |

| Group (n=10) | Subcutaneous Treatment | Mean Withdrawal Response (g) - Day 3 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Predose | | 30 min pd | | 60 min pd | | 90 min pd | | 120 min pd | |
| | | Left | Right | Left | Right | Left | Right | Left | Right | Left | Right |
| 8 | Morphine sulfate 3 mg/kg 1 mL./kg | 9.7 ± 0.75 | 3.6 ± 0.43 | 33.1* ± 5.08 | 17.9* ± 1.72 | 28.5* ± 5.00 | 19.3* ± 3.93 | 18.4 ± 4.80 | 7.6* ± 1.40 | 13.1 ± 2.83 | 5.6 ± 0.84 |

| Group (n=10) | Subcutaneous Treatment | Mean Withdrawal Response (g) - Day 6 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Predose | | 30 min pd | | 60 min pd | | 90 min pd | | 120 min pd | |
| | | Left | Right | Left | Right | Left | Right | Left | Right | Left | Right |
| 8 | Morphine sulfate 3 mg/kg 1 mL./kg | 8.7 ± 1.01 | 3.5 ± 0.33 | 38.5* ± 5.10 | 22.8* ± 2.87 | 27.9* ± 4.66 | 16.9* ± 1.43 | 17.2* ± 2.52 | 10.1* ± 1.40 | 13.5* ± 2.10 | 6.9* ± 1.34 |

Data presented as mean ± standard error of the mean

[a] Doses were administered on the day of surgery (1 hour before surgery for Groups 1 and 4 and 3 hours after surgery for Groups 2, 3 and 5-8) and on Days 3 and 6 after the behavioral testing.

* Statistically significant (p≤0.05) change compared to the predose values – Student's paired t-test.

# *XI. Appendices*

# A. Appendix I - Individual Animal Data

### *Individual Animal Data – Mean Withdrawal Response (g) - Pre-surgery*

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 15 | 8 | 15 | 12.7 | 15 | 8 | 15 | 12.7 |
| | 1702 | | 15 | 15 | 15 | 15.0 | 26 | 15 | 8 | 16.3 |
| | 1703 | | 15 | 8 | 26 | 16.3 | 26 | 15 | 8 | 16.3 |
| | 1704 | | 8 | 15 | 8 | 10.3 | 15 | 15 | 15 | 15.0 |
| | 1705 | | 15 | 8 | 8 | 10.3 | 26 | 8 | 8 | 14.0 |
| | 1706 | | 8 | 15 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1707 | | 15 | 15 | 8 | 12.7 | 26 | 8 | 15 | 16.3 |
| | 1708 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 15 | 12.7 |
| | 1709 | | 15 | 15 | 15 | 15.0 | 26 | 15 | 8 | 16.3 |
| | 1710 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 15 | 12.7 |
| 2 | 1711 | Treatment B | 15 | 15 | 15 | 15.0 | 15 | 8 | 15 | 12.7 |
| | 1712 | | 26 | 15 | 26 | 22.3 | 15 | 15 | 15 | 15.0 |
| | 1713 | | 15 | 15 | 15 | 15.0 | 15 | 26 | 8 | 16.3 |
| | 1714 | | 8 | 15 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1715 | | 8 | 15 | 26 | 16.3 | 15 | 15 | 8 | 12.7 |
| | 1716 | | 26 | 26 | 26 | 26.0 | 15 | 15 | 26 | 18.7 |
| | 1717 | | 8 | 8 | 15 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1718 | | 26 | 15 | 15 | 18.7 | 26 | 15 | 15 | 18.7 |
| | 1719 | | 8 | 15 | 8 | 10.3 | 15 | 15 | 15 | 15.0 |
| | 1720 | | 15 | 15 | 15 | 15.0 | 8 | 15 | 8 | 10.3 |
| 3 | 1721 | Treatment C | 15 | 15 | 15 | 15.0 | 15 | 8 | 8 | 10.3 |
| | 1722 | | 8 | 15 | 8 | 10.3 | 15 | 8 | 8 | 10.3 |
| | 1723 | | 15 | 15 | 15 | 15.0 | 15 | 15 | 15 | 15.0 |
| | 1781 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 15 | 12.7 |
| | 1725 | | 26 | 15 | 8 | 16.3 | 15 | 15 | 15 | 15.0 |
| | 1726 | | 15 | 15 | 8 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1727 | | 15 | 15 | 15 | 15.0 | 15 | 15 | 8 | 12.7 |
| | 1728 | | 26 | 15 | 15 | 18.7 | 15 | 15 | 8 | 12.7 |
| | 1729 | | 8 | 15 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1730 | | 15 | 8 | 8 | 10.3 | 15 | 8 | 15 | 12.7 |

*Individual Animal Data – Mean Withdrawal Response (g) - Pre-surgery*

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1731 | Treatment D | 15 | 6 | 8 | 9.7 | 26 | 15 | 8 | 16.3 |
| | 1732 | | 15 | 15 | 15 | 15.0 | 15 | 15 | 15 | 15.0 |
| | 1733 | | 15 | 15 | 8 | 12.7 | 15 | 8 | 15 | 12.7 |
| | 1734 | | 26 | 26 | 15 | 22.3 | 15 | 26 | 8 | 16.3 |
| | 1735 | | 15 | 15 | 8 | 12.7 | 15 | 15 | 15 | 15.0 |
| | 1736 | | 15 | 8 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1737 | | 26 | 26 | 15 | 22.3 | 26 | 26 | 26 | 26.0 |
| | 1738 | | 15 | 8 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1739 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1740 | | 15 | 15 | 8 | 12.7 | 15 | 15 | 15 | 15.0 |
| 5 | 1741 | Treatment E | 15 | 8 | 8 | 10.3 | 26 | 8 | 15 | 16.3 |
| | 1742 | | 15 | 15 | 15 | 15.0 | 26 | 26 | 15 | 22.3 |
| | 1743 | | 26 | 26 | 15 | 22.3 | 8 | 26 | 15 | 16.3 |
| | 1744 | | 15 | 8 | 15 | 12.7 | 26 | 8 | 8 | 14.0 |
| | 1745 | | 26 | 15 | 15 | 18.7 | 26 | 8 | 15 | 16.3 |
| | 1746 | | 15 | 15 | 8 | 12.7 | 26 | 8 | 15 | 16.3 |
| | 1747 | | 8 | 8 | 15 | 10.3 | 26 | 8 | 15 | 16.3 |
| | 1748 | | 15 | 8 | 8 | 10.3 | 8 | 8 | 15 | 10.3 |
| | 1749 | | 8 | 15 | 8 | 10.3 | 8 | 8 | 15 | 10.3 |
| | 1750 | | 26 | 15 | 26 | 22.3 | 26 | 15 | 8 | 16.3 |
| 6 | 1751 | Treatment F | 15 | 8 | 15 | 12.7 | 26 | 15 | 15 | 18.7 |
| | 1752 | | 15 | 8 | 8 | 10.3 | 15 | 8 | 15 | 12.7 |
| | 1753 | | 8 | 15 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1754 | | 15 | 15 | 15 | 15.0 | 8 | 15 | 15 | 12.7 |
| | 1755 | | 8 | 15 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1756 | | 26 | 26 | 26 | 26.0 | 26 | 15 | 26 | 22.3 |
| | 1757 | | 15 | 8 | 8 | 10.3 | 15 | 8 | 8 | 10.3 |
| | 1758 | | 15 | 6 | 15 | 12.0 | 15 | 15 | 8 | 12.7 |
| | 1759 | | 26 | 15 | 15 | 18.7 | 26 | 15 | 15 | 18.7 |
| | 1760 | | 15 | 8 | 26 | 16.3 | 15 | 8 | 8 | 10.3 |

***Individual Animal Data – Mean Withdrawal Response (g) - Pre-surgery***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 8 | 15 | 15 | 12.7 | 15 | 15 | 8 | 12.7 |
| | 1762 | | 15 | 15 | 15 | 15.0 | 15 | 15 | 8 | 12.7 |
| | 1763 | | 6 | 15 | 15 | 12.0 | 15 | 8 | 15 | 12.7 |
| | 1765 | | 15 | 8 | 8 | 10.3 | 15 | 15 | 8 | 12.7 |
| | 1766 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 15 | 12.7 |
| | 1767 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1768 | | 15 | 15 | 8 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1769 | | 26 | 26 | 26 | 26.0 | 15 | 15 | 15 | 15.0 |
| | 1770 | | 26 | 15 | 15 | 18.7 | 26 | 15 | 15 | 18.7 |
| | 1782 | | 26 | 8 | 6 | 13.3 | 26 | 15 | 8 | 16.3 |
| 8 | 1771 | Morphine sulfate | 26 | 15 | 15 | 18.7 | 26 | 26 | 26 | 26.0 |
| | 1772 | | 15 | 26 | 26 | 22.3 | 15 | 26 | 15 | 18.7 |
| | 1773 | | 26 | 8 | 15 | 16.3 | 8 | 8 | 15 | 10.3 |
| | 1774 | | 15 | 8 | 15 | 12.7 | 26 | 8 | 15 | 16.3 |
| | 1775 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 8 | 10.3 |
| | 1776 | | 15 | 15 | 15 | 15.0 | 15 | 26 | 8 | 16.3 |
| | 1777 | | 26 | 26 | 26 | 26.0 | 15 | 15 | 8 | 12.7 |
| | 1778 | | 15 | 26 | 15 | 18.7 | 26 | 15 | 15 | 18.7 |
| | 1779 | | 8 | 15 | 15 | 12.7 | 26 | 26 | 8 | 20.0 |
| | 1780 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 0 (2 Hours Post-Surgery)***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 6 | 6 | 15 | 9.0 | 4 | 4 | 6 | 4.7 |
| | 1702 | | 8 | 6 | 8 | 7.3 | 15 | 2 | 2 | 6.3 |
| | 1703 | | 6 | 6 | 8 | 6.7 | 4 | 1.4 | 1 | 2.1 |
| | 1704 | | 15 | 26 | 8 | 16.3 | 2 | 4 | 2 | 2.7 |
| | 1705 | | 26 | 26 | 15 | 22.3 | 2 | 4 | 4 | 3.3 |
| | 1706 | | 26 | 26 | 26 | 26.0 | 4 | 2 | 4 | 3.3 |
| | 1707 | | 26 | 26 | 15 | 22.3 | 2 | 1.4 | 1.4 | 1.6 |
| | 1708 | | 15 | 26 | 26 | 22.3 | 4 | 4 | 4 | 4.0 |
| | 1709 | | 6 | 15 | 8 | 9.7 | 4 | 2 | 1.4 | 2.5 |
| | 1710 | | 26 | 26 | 15 | 22.3 | 4 | 2 | 2 | 2.7 |
| 2 | 1711 | Treatment B | 26 | 26 | 15 | 22.3 | 26 | 26 | 15 | 22.3 |
| | 1712 | | 15 | 15 | 15 | 15.0 | 8 | 8 | 8 | 8.0 |
| | 1713 | | 26 | 8 | 8 | 14.0 | 26 | 15 | 15 | 18.7 |
| | 1714 | | 26 | 26 | 26 | 26.0 | 26 | 15 | 15 | 18.7 |
| | 1715 | | 26 | 26 | 15 | 22.3 | 15 | 15 | 8 | 12.7 |
| | 1716 | | 26 | 26 | 15 | 22.3 | 26 | 26 | 15 | 22.3 |
| | 1717 | | 15 | 8 | 6 | 9.7 | 15 | 15 | 8 | 12.7 |
| | 1718 | | 15 | 8 | 8 | 10.3 | 15 | 8 | 15 | 12.7 |
| | 1719 | | 15 | 6 | 8 | 9.7 | 6 | 6 | 8 | 6.7 |
| | 1720 | | 8 | 15 | 8 | 10.3 | 15 | 8 | 8 | 10.3 |
| 3 | 1721 | Treatment C | 15 | 8 | 8 | 10.3 | 2 | 1 | 1 | 1.3 |
| | 1722 | | 26 | 8 | 15 | 16.3 | 2 | 2 | 1.4 | 1.8 |
| | 1723 | | 4 | 4 | 8 | 5.3 | 1.4 | 2 | 4 | 2.5 |
| | 1781 | | 15 | 15 | 8 | 12.7 | 2 | 4 | 2 | 2.7 |
| | 1725 | | 15 | 15 | 6 | 12.0 | 0.6 | 2 | 2 | 1.5 |
| | 1726 | | 15 | 15 | 15 | 15.0 | 4 | 2 | 1 | 2.3 |
| | 1727 | | 8 | 15 | 15 | 12.7 | 1.4 | 1 | 1.4 | 1.3 |
| | 1728 | | 26 | 15 | 26 | 22.3 | 4 | 4 | 4 | 4.0 |
| | 1729 | | 8 | 15 | 8 | 10.3 | 6 | 4 | 4 | 4.7 |
| | 1730 | | 15 | 6 | 15 | 12.0 | 6 | 4 | 1.4 | 3.8 |

### *Individual Animal Data – Mean Withdrawal Response (g) – Day 0 (2 Hours Post-Surgery)*

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1731 | Treatment D | 15 | 26 | 8 | 16.3 | 4 | 6 | 4 | 4.7 |
| | 1732 | | 15 | 6 | 8 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1733 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1734 | | 15 | 8 | 8 | 10.3 | 2 | 1.4 | 1.4 | 1.6 |
| | 1735 | | 26 | 8 | 15 | 16.3 | 6 | 1.4 | 2 | 3.1 |
| | 1736 | | 15 | 15 | 8 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1737 | | 15 | 15 | 15 | 15.0 | 4 | 6 | 4 | 4.7 |
| | 1738 | | 8 | 15 | 8 | 10.3 | 4 | 4 | 2 | 3.3 |
| | 1739 | | 8 | 15 | 6 | 9.7 | 2 | 1 | 1 | 1.3 |
| | 1740 | | 26 | 26 | 15 | 22.3 | 4 | 2 | 2 | 2.7 |
| 5 | 1741 | Treatment E | 15 | 8 | 15 | 12.7 | 2 | 2 | 2 | 2.0 |
| | 1742 | | 15 | 8 | 26 | 16.3 | 4 | 6 | 6 | 5.3 |
| | 1743 | | 15 | 15 | 26 | 18.7 | 4 | 2 | 2 | 2.7 |
| | 1744 | | 26 | 8 | 8 | 14.0 | 4 | 2 | 2 | 2.7 |
| | 1745 | | 15 | 15 | 15 | 15.0 | 4 | 2 | 2 | 2.7 |
| | 1746 | | 26 | 15 | 15 | 18.7 | 4 | 4 | 1 | 3.0 |
| | 1747 | | 26 | 26 | 15 | 22.3 | 4 | 4 | 6 | 4.7 |
| | 1748 | | 26 | 8 | 15 | 16.3 | 1.4 | 1 | 1.4 | 1.3 |
| | 1749 | | 8 | 8 | 8 | 8.0 | 6 | 4 | 1.4 | 3.8 |
| | 1750 | | 15 | 26 | 26 | 22.3 | 1 | 4 | 2 | 2.3 |
| 6 | 1751 | Treatment F | 26 | 8 | 26 | 20.0 | 1.4 | 4 | 1.4 | 2.3 |
| | 1752 | | 15 | 8 | 15 | 12.7 | 4 | 2 | 6 | 4.0 |
| | 1753 | | 15 | 15 | 15 | 15.0 | 2 | 4 | 4 | 3.3 |
| | 1754 | | 26 | 26 | 15 | 22.3 | 8 | 4 | 4 | 5.3 |
| | 1755 | | 15 | 8 | 26 | 16.3 | 4 | 2 | 4 | 3.3 |
| | 1756 | | 15 | 26 | 26 | 22.3 | 4 | 2 | 2 | 2.7 |
| | 1757 | | 26 | 15 | 8 | 16.3 | 6 | 2 | 2 | 3.3 |
| | 1758 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 1 | 3.7 |
| | 1759 | | 8 | 6 | 8 | 7.3 | 4 | 1.4 | 2 | 2.5 |
| | 1760 | | 15 | 26 | 26 | 22.3 | 4 | 4 | 4 | 4.0 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 0 (2 Hours Post-Surgery)***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 26 | 8 | 8 | 14.0 | 6 | 6 | 4 | 26 |
| | 1762 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 2 | 15 |
| | 1763 | | 8 | 8 | 15 | 10.3 | 2 | 4 | 2 | 8 |
| | 1765 | | 8 | 15 | 15 | 12.7 | 4 | 4 | 4 | 8 |
| | 1766 | | 15 | 15 | 8 | 12.7 | 8 | 8 | 6 | 15 |
| | 1767 | | 15 | 8 | 15 | 12.7 | 2 | 1.4 | 1 | 15 |
| | 1768 | | 8 | 6 | 15 | 9.7 | 6 | 2 | 1.4 | 8 |
| | 1769 | | 8 | 8 | 15 | 10.3 | 1.4 | 1 | 1 | 8 |
| | 1770 | | 15 | 8 | 8 | 10.3 | 4 | 2 | 1 | 15 |
| | 1782 | | 15 | 15 | 15 | 15.0 | 2 | 1.4 | 1.4 | 15 |
| 8 | 1771 | Morphine sulfate | 15 | 6 | 8 | 9.7 | 4 | 2 | 1.4 | 15 |
| | 1772 | | 15 | 15 | 8 | 12.7 | 4 | 4 | 4 | 15 |
| | 1773 | | 15 | 8 | 6 | 9.7 | 6 | 4 | 4 | 15 |
| | 1774 | | 8 | 15 | 6 | 9.7 | 2 | 2 | 4 | 8 |
| | 1775 | | 26 | 26 | 15 | 22.3 | 6 | 4 | 4 | 26 |
| | 1776 | | 8 | 8 | 8 | 8.0 | 4 | 4 | 2 | 8 |
| | 1777 | | 26 | 15 | 8 | 16.3 | 4 | 2 | 4 | 26 |
| | 1778 | | 15 | 26 | 8 | 16.3 | 4 | 4 | 4 | 15 |
| | 1779 | | 8 | 15 | 15 | 12.7 | 2 | 1 | 1 | 8 |
| | 1780 | | 15 | 15 | 15 | 15.0 | 4 | 4 | 2 | 15 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 1***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 8 | 15 | 15 | 12.7 | 6 | 4 | 4 | 4.7 |
| | 1702 | | 26 | 8 | 8 | 14.0 | 6 | 4 | 6 | 5.3 |
| | 1703 | | 8 | 15 | 15 | 12.7 | 4 | 2 | 2 | 2.7 |
| | 1704 | | 15 | 15 | 8 | 12.7 | 8 | 8 | 4 | 6.7 |
| | 1705 | | 26 | 15 | 15 | 18.7 | 15 | 15 | 8 | 12.7 |
| | 1706 | | 8 | 15 | 8 | 10.3 | 4 | 4 | 2 | 3.3 |
| | 1707 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 4 | 4.7 |
| | 1708 | | 15 | 15 | 15 | 15.0 | 6 | 4 | 2 | 4.0 |
| | 1709 | | 8 | 8 | 15 | 10.3 | 4 | 2 | 4 | 3.3 |
| | 1710 | | 15 | 15 | 15 | 15.0 | 4 | 4 | 6 | 4.7 |
| 2 | 1711 | Treatment B | 15 | 8 | 15 | 12.7 | 15 | 15 | 8 | 12.7 |
| | 1712 | | 26 | 8 | 15 | 16.3 | 15 | 8 | 8 | 10.3 |
| | 1713 | | 8 | 15 | 6 | 9.7 | 26 | 8 | 6 | 13.3 |
| | 1714 | | 15 | 15 | 26 | 18.7 | 26 | 15 | 15 | 18.7 |
| | 1715 | | 15 | 15 | 15 | 15.0 | 8 | 15 | 15 | 12.7 |
| | 1716 | | 15 | 26 | 15 | 18.7 | 15 | 26 | 15 | 18.7 |
| | 1717 | | 26 | 15 | 8 | 16.3 | 15 | 15 | 15 | 15.0 |
| | 1718 | | 15 | 15 | 8 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1719 | | 8 | 6 | 15 | 9.7 | 8 | 6 | 15 | 9.7 |
| | 1720 | | 8 | 8 | 15 | 10.3 | 15 | 8 | 15 | 12.7 |
| 3 | 1721 | Treatment C | 15 | 15 | 6 | 12.0 | 1.4 | 1 | 1 | 1.1 |
| | 1722 | | 26 | 6 | 6 | 12.7 | 6 | 6 | 6 | 6.0 |
| | 1723 | | 8 | 15 | 6 | 9.7 | 1.4 | 1 | 1 | 1.1 |
| | 1781 | | 15 | 8 | 8 | 10.3 | 4 | 1.4 | 1.4 | 2.3 |
| | 1725 | | 15 | 15 | 15 | 15.0 | 4 | 4 | 2 | 3.3 |
| | 1726 | | 15 | 26 | 15 | 18.7 | 4 | 4 | 2 | 3.3 |
| | 1727 | | 15 | 26 | 15 | 18.7 | 1 | 1.4 | 1 | 1.1 |
| | 1728 | | 8 | 6 | 8 | 7.3 | 6 | 4 | 4 | 4.7 |
| | 1729 | | 26 | 15 | 15 | 18.7 | 4 | 2 | 1.4 | 2.5 |
| | 1730 | | 8 | 15 | 8 | 10.3 | 4 | 4 | 4 | 4.0 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 1***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1731 | Treatment D | 15 | 15 | 15 | 15.0 | 4 | 6 | 6 | 5.3 |
| | 1732 | | 6 | 4 | 8 | 6.0 | 0.6 | 2 | 4 | 2.2 |
| | 1733 | | 8 | 8 | 8 | 8.0 | 1 | 1 | 4 | 2.0 |
| | 1734 | | 15 | 6 | 6 | 9.0 | 4 | 1.4 | 1.4 | 2.3 |
| | 1735 | | 8 | 15 | 8 | 10.3 | 6 | 4 | 2 | 4.0 |
| | 1736 | | 15 | 8 | 15 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1737 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 4 | 4.7 |
| | 1738 | | 8 | 8 | 8 | 8.0 | 4 | 2 | 2 | 2.7 |
| | 1739 | | 8 | 8 | 8 | 8.0 | 2 | 2 | 2 | 2.0 |
| | 1740 | | 8 | 8 | 15 | 10.3 | 6 | 6 | 4 | 5.3 |
| 5 | 1741 | Treatment E | 15 | 8 | 15 | 12.7 | 4 | 4 | 4 | 4.0 |
| | 1742 | | 15 | 26 | 26 | 22.3 | 15 | 6 | 6 | 9.0 |
| | 1743 | | 8 | 15 | 15 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1744 | | 8 | 6 | 15 | 9.7 | 4 | 1.4 | 1 | 2.1 |
| | 1745 | | 15 | 6 | 6 | 9.0 | 6 | 4 | 6 | 5.3 |
| | 1746 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 2 | 4.0 |
| | 1747 | | 8 | 15 | 8 | 10.3 | 6 | 6 | 2 | 4.7 |
| | 1748 | | 15 | 8 | 8 | 10.3 | 4 | 2 | 4 | 3.3 |
| | 1749 | | 26 | 8 | 8 | 14.0 | 4 | 4 | 2 | 3.3 |
| | 1750 | | 15 | 15 | 8 | 12.7 | 2 | 1.4 | 4 | 2.5 |
| 6 | 1751 | Treatment F | 26 | 15 | 15 | 18.7 | 6 | 2 | 1.4 | 3.1 |
| | 1752 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 6 | 5.3 |
| | 1753 | | 15 | 8 | 15 | 12.7 | 4 | 4 | 4 | 4.0 |
| | 1754 | | 4 | 4 | 4 | 4.0 | 2 | 4 | 2 | 2.7 |
| | 1755 | | 15 | 6 | 8 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1756 | | 8 | 15 | 15 | 12.7 | 6 | 2 | 2 | 3.3 |
| | 1757 | | 15 | 15 | 26 | 18.7 | 4 | 2 | 2 | 2.7 |
| | 1758 | | 8 | 15 | 8 | 10.3 | 4 | 2 | 4 | 3.3 |
| | 1759 | | 8 | 6 | 8 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1760 | | 26 | 15 | 15 | 18.7 | 8 | 4 | 4 | 5.3 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 1***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 8 | 8 | 6 | 7.3 | 6 | 4 | 6 | 5.3 |
| | 1762 | | 15 | 6 | 8 | 9.7 | 6 | 6 | 4 | 5.3 |
| | 1763 | | 15 | 8 | 8 | 10.3 | 6 | 2 | 6 | 4.7 |
| | 1765 | | 15 | 8 | 15 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1766 | | 15 | 6 | 15 | 12.0 | 4 | 1.4 | 2 | 2.5 |
| | 1767 | | 8 | 15 | 8 | 10.3 | 4 | 4 | 1.4 | 3.1 |
| | 1768 | | 15 | 15 | 8 | 12.7 | 4 | 2 | 1.4 | 2.5 |
| | 1769 | | 15 | 15 | 8 | 12.7 | 4 | 2 | 2 | 2.7 |
| | 1770 | | 15 | 15 | 8 | 12.7 | 4 | 4 | 4 | 4.0 |
| | 1782 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 4 | 4.7 |
| 8 | 1771 | Morphine sulfate | 15 | 15 | 8 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1772 | | 6 | 8 | 8 | 7.3 | 1.4 | 1 | 2 | 1.5 |
| | 1773 | | 8 | 8 | 6 | 7.3 | 4 | 2 | 2 | 2.7 |
| | 1774 | | 15 | 8 | 15 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1775 | | 15 | 8 | 8 | 10.3 | 6 | 2 | 4 | 4.0 |
| | 1776 | | 15 | 26 | 8 | 16.3 | 6 | 4 | 4 | 4.7 |
| | 1777 | | 15 | 8 | 15 | 12.7 | 2 | 1.4 | 4 | 2.5 |
| | 1778 | | 26 | 15 | 15 | 18.7 | 4 | 4 | 4 | 4.0 |
| | 1779 | | 15 | 8 | 8 | 10.3 | 4 | 1.4 | 1.4 | 2.3 |
| | 1780 | | 8 | 8 | 8 | 8.0 | 4 | 2 | 1 | 2.3 |

### *Individual Animal Data – Mean Withdrawal Response (g) – Day 3*

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 6 | 8 | 15 | 9.7 | 6 | 4 | 2 | 4.0 |
| | 1702 | | 8 | 15 | 8 | 10.3 | 2 | 4 | 6 | 4.0 |
| | 1703 | | 8 | 8 | 15 | 10.3 | 4 | 6 | 8 | 6.0 |
| | 1704 | | 15 | 26 | 8 | 16.3 | 4 | 4 | 2 | 3.3 |
| | 1705 | | 15 | 15 | 15 | 15.0 | 6 | 8 | 2 | 5.3 |
| | 1706 | | 15 | 15 | 15 | 15.0 | 2 | 1.4 | 4 | 2.5 |
| | 1707 | | 8 | 8 | 8 | 8.0 | 6 | 6 | 2 | 4.7 |
| | 1708 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 4 | 4.7 |
| | 1709 | | 15 | 8 | 8 | 10.3 | 2 | 4 | 2 | 2.7 |
| | 1710 | | 26 | 8 | 8 | 14.0 | 15 | 4 | 4 | 7.7 |
| 2 | 1711 | Treatment B | 8 | 8 | 6 | 7.3 | 6 | 6 | 8 | 6.7 |
| | 1712 | | 15 | 15 | 8 | 12.7 | 15 | 15 | 15 | 15.0 |
| | 1713 | | 8 | 6 | 8 | 7.3 | 15 | 26 | 15 | 18.7 |
| | 1714 | | 15 | 26 | 15 | 18.7 | 15 | 26 | 15 | 18.7 |
| | 1715 | | 15 | 15 | 8 | 12.7 | 26 | 26 | 15 | 22.3 |
| | 1716 | | 15 | 15 | 8 | 12.7 | 6 | 15 | 15 | 12.0 |
| | 1717 | | 15 | 26 | 8 | 16.3 | 6 | 8 | 15 | 9.7 |
| | 1718 | | 8 | 8 | 8 | 8.0 | 8 | 6 | 8 | 7.3 |
| | 1719 | | 8 | 8 | 6 | 7.3 | 15 | 8 | 8 | 10.3 |
| | 1720 | | 15 | 15 | 6 | 12.0 | 15 | 8 | 15 | 12.7 |
| 3 | 1721 | Treatment C | 15 | 15 | 15 | 15.0 | 8 | 8 | 2 | 6.0 |
| | 1722 | | 15 | 15 | 8 | 12.7 | 6 | 8 | 4 | 6.0 |
| | 1723 | | 15 | 8 | 15 | 12.7 | 15 | 8 | 8 | 10.3 |
| | 1781 | | 15 | 15 | 6 | 12.0 | 4 | 4 | 6 | 4.7 |
| | 1725 | | 15 | 15 | 8 | 12.7 | 8 | 15 | 4 | 9.0 |
| | 1726 | | 15 | 8 | 8 | 10.3 | 6 | 6 | 6 | 6.0 |
| | 1727 | | 15 | 8 | 8 | 10.3 | 2 | 6 | 4 | 4.0 |
| | 1728 | | 8 | 8 | 8 | 8.0 | 8 | 6 | 2 | 5.3 |
| | 1729 | | 15 | 8 | 8 | 10.3 | 6 | 6 | 4 | 5.3 |
| | 1730 | | 8 | 8 | 15 | 10.3 | 6 | 4 | 2 | 4.0 |

### *Individual Animal Data – Mean Withdrawal Response (g) – Day 3*

| | | | Left Paw | | | | Right Paw | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Group Number** | **Ear Tag Number** | **Treatment** | **Reading 1** | **Reading 2** | **Reading 3** | **Mean** | **Reading 1** | **Reading 2** | **Reading 3** | **Mean** |
| 4 | 1731 | Treatment D | 15 | 8 | 8 | 10.3 | 6 | 8 | 15 | 9.7 |
| | 1732 | | 15 | 26 | 8 | 16.3 | 6 | 6 | 6 | 6.0 |
| | 1733 | | 15 | 15 | 8 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1734 | | 15 | 8 | 6 | 9.7 | 6 | 6 | 6 | 6.0 |
| | 1735 | | 8 | 15 | 8 | 10.3 | 6 | 4 | 4 | 4.7 |
| | 1736 | | 15 | 8 | 8 | 10.3 | 4 | 4 | 4 | 4.0 |
| | 1737 | | 15 | 15 | 15 | 15.0 | 4 | 6 | 4 | 4.7 |
| | 1738 | | 8 | 6 | 15 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1739 | | 8 | 6 | 8 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1740 | | 8 | 6 | 8 | 7.3 | 2 | 6 | 4 | 4.0 |
| 5 | 1741 | Treatment E | 15 | 15 | 8 | 12.7 | 8 | 6 | 4 | 6.0 |
| | 1742 | | 15 | 15 | 8 | 12.7 | 8 | 4 | 6 | 6.0 |
| | 1743 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 8 | 10.3 |
| | 1744 | | 8 | 6 | 15 | 9.7 | 4 | 6 | 6 | 5.3 |
| | 1745 | | 8 | 15 | 8 | 10.3 | 6 | 6 | 8 | 6.7 |
| | 1746 | | 15 | 15 | 8 | 12.7 | 8 | 2 | 1.4 | 3.8 |
| | 1747 | | 15 | 26 | 26 | 22.3 | 8 | 6 | 2 | 5.3 |
| | 1748 | | 8 | 15 | 8 | 10.3 | 4 | 6 | 2 | 4.0 |
| | 1749 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1750 | | 15 | 8 | 6 | 9.7 | 15 | 4 | 4 | 7.7 |
| 6 | 1751 | Treatment F | 26 | 15 | 8 | 16.3 | 8 | 6 | 4 | 6.0 |
| | 1752 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 4 | 9.0 |
| | 1753 | | 15 | 15 | 15 | 15.0 | 6 | 4 | 4 | 4.7 |
| | 1754 | | 8 | 8 | 6 | 7.3 | 15 | 6 | 6 | 9.0 |
| | 1755 | | 15 | 15 | 15 | 15.0 | 15 | 4 | 4 | 7.7 |
| | 1756 | | 15 | 8 | 8 | 10.3 | 4 | 4 | 2 | 3.3 |
| | 1757 | | 15 | 15 | 6 | 12.0 | 6 | 8 | 8 | 7.3 |
| | 1758 | | 8 | 15 | 6 | 9.7 | 6 | 4 | 8 | 6.0 |
| | 1759 | | 15 | 8 | 8 | 10.3 | 4 | 4 | 4 | 4.0 |
| | 1760 | | 8 | 15 | 15 | 12.7 | 4 | 4 | 4 | 4.0 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 3***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 15 | 8 | 6 | 9.7 | 8 | 4 | 6 | 6.0 |
| | 1762 | | 8 | 6 | 6 | 6.7 | 8 | 6 | 4 | 6.0 |
| | 1763 | | 8 | 6 | 6 | 6.7 | 8 | 6 | 4 | 6.0 |
| | 1765 | | 26 | 15 | 15 | 18.7 | 8 | 4 | 6 | 6.0 |
| | 1766 | | 15 | 8 | 6 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1767 | | 26 | 15 | 8 | 16.3 | 2 | 4 | 2 | 2.7 |
| | 1768 | | 8 | 8 | 15 | 10.3 | 6 | 4 | 4 | 4.7 |
| | 1769 | | 15 | 8 | 15 | 12.7 | 2 | 2 | 1.4 | 1.8 |
| | 1770 | | 15 | 15 | 8 | 12.7 | 4 | 2 | 6 | 4.0 |
| | 1782 | | 8 | 8 | 6 | 7.3 | 6 | 4 | 4 | 4.7 |
| 8 | 1771 | Morphine sulfate | 8 | 6 | 6 | 6.7 | 4 | 4 | 4 | 4.0 |
| | 1772 | | 8 | 8 | 8 | 8.0 | 6 | 4 | 6 | 5.3 |
| | 1773 | | 6 | 8 | 8 | 7.3 | 4 | 4 | 6 | 4.7 |
| | 1774 | | 8 | 8 | 8 | 8.0 | 2 | 4 | 4 | 3.3 |
| | 1775 | | 15 | 8 | 8 | 10.3 | 2 | 2 | 4 | 2.7 |
| | 1776 | | 8 | 15 | 8 | 10.3 | 2 | 1.4 | 1.4 | 1.6 |
| | 1777 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 2 | 4.0 |
| | 1778 | | 15 | 15 | 15 | 15.0 | 4 | 6 | 4 | 4.7 |
| | 1779 | | 15 | 8 | 8 | 10.3 | 1.4 | 1.4 | 1.4 | 1.4 |
| | 1780 | | 8 | 15 | 8 | 10.3 | 6 | 6 | 2 | 4.7 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 6***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 8 | 8 | 8 | 8.0 | 15 | 6 | 6 | 9.0 |
| | 1702 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 4 | 4.7 |
| | 1703 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1704 | | 6 | 6 | 4 | 5.3 | 8 | 8 | 6 | 7.3 |
| | 1705 | | 8 | 8 | 8 | 8.0 | 8 | 6 | 6 | 6.7 |
| | 1706 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 4 | 4.7 |
| | 1707 | | 15 | 15 | 8 | 12.7 | 6 | 6 | 2 | 4.7 |
| | 1708 | | 26 | 15 | 8 | 16.3 | 15 | 4 | 6 | 8.3 |
| | 1709 | | 8 | 8 | 6 | 7.3 | 6 | 4 | 6 | 5.3 |
| | 1710 | | 8 | 6 | 8 | 7.3 | 6 | 4 | 6 | 5.3 |
| 2 | 1711 | Treatment B | 15 | 15 | 8 | 12.7 | 15 | 8 | 6 | 9.7 |
| | 1712 | | 15 | 6 | 6 | 9.0 | 15 | 15 | 8 | 12.7 |
| | 1713 | | 8 | 6 | 6 | 6.7 | 15 | 4 | 2 | 7.0 |
| | 1714 | | 15 | 8 | 8 | 10.3 | 8 | 6 | 4 | 6.0 |
| | 1715 | | 8 | 4 | 4 | 5.3 | 15 | 8 | 8 | 10.3 |
| | 1716 | | 15 | 15 | 15 | 15.0 | 8 | 15 | 15 | 12.7 |
| | 1717 | | 15 | 8 | 15 | 12.7 | 26 | 8 | 15 | 16.3 |
| | 1718 | | 15 | 26 | 8 | 16.3 | 8 | 8 | 8 | 8.0 |
| | 1719 | | 15 | 15 | 15 | 15.0 | 15 | 8 | 6 | 9.7 |
| | 1720 | | 8 | 8 | 8 | 8.0 | 8 | 15 | 15 | 12.7 |
| 3 | 1721 | Treatment C | 8 | 15 | 8 | 10.3 | 8 | 4 | 4 | 5.3 |
| | 1722 | | 6 | 8 | 15 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1723 | | 8 | 8 | 15 | 10.3 | 15 | 8 | 6 | 9.7 |
| | 1781 | | 15 | 8 | 15 | 12.7 | 4 | 6 | 8 | 6.0 |
| | 1725 | | 8 | 15 | 15 | 12.7 | 8 | 8 | 8 | 8.0 |
| | 1726 | | 8 | 26 | 8 | 14.0 | 4 | 4 | 8 | 5.3 |
| | 1727 | | 15 | 6 | 6 | 9.0 | 8 | 4 | 2 | 4.7 |
| | 1728 | | 8 | 8 | 6 | 7.3 | 8 | 6 | 4 | 6.0 |
| | 1729 | | 15 | 8 | 26 | 16.3 | 4 | 8 | 4 | 5.3 |
| | 1730 | | 8 | 15 | 15 | 12.7 | 8 | 4 | 8 | 6.7 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 6***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1731 | Treatment D | 15 | 8 | 8 | 10.3 | 15 | 15 | 6 | 12.0 |
| | 1732 | | 8 | 6 | 4 | 6.0 | 15 | 6 | 6 | 9.0 |
| | 1733 | | 6 | 4 | 4 | 4.7 | 6 | 6 | 4 | 5.3 |
| | 1734 | | 8 | 6 | 6 | 6.7 | 8 | 6 | 4 | 6.0 |
| | 1735 | | 8 | 8 | 6 | 7.3 | 4 | 6 | 4 | 4.7 |
| | 1736 | | 15 | 8 | 8 | 10.3 | 6 | 4 | 6 | 5.3 |
| | 1737 | | 26 | 15 | 15 | 18.7 | 4 | 6 | 4 | 4.7 |
| | 1738 | | 15 | 8 | 15 | 12.7 | 4 | 4 | 4 | 4.0 |
| | 1739 | | 8 | 6 | 8 | 7.3 | 6 | 6 | 2 | 4.7 |
| | 1740 | | 15 | 8 | 6 | 9.7 | 4 | 6 | 4 | 4.7 |
| 5 | 1741 | Treatment E | 15 | 8 | 8 | 10.3 | 8 | 8 | 6 | 7.3 |
| | 1742 | | 15 | 15 | 8 | 12.7 | 8 | 8 | 8 | 8.0 |
| | 1743 | | 8 | 8 | 8 | 8.0 | 8 | 6 | 6 | 6.7 |
| | 1744 | | 6 | 4 | 4 | 4.7 | 8 | 6 | 8 | 7.3 |
| | 1745 | | 15 | 15 | 8 | 12.7 | 6 | 6 | 6 | 6.0 |
| | 1746 | | 15 | 8 | 8 | 10.3 | 4 | 2 | 4 | 3.3 |
| | 1747 | | 15 | 15 | 8 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1748 | | 8 | 8 | 6 | 7.3 | 4 | 2 | 8 | 4.7 |
| | 1749 | | 15 | 15 | 15 | 15.0 | 4 | 6 | 4 | 4.7 |
| | 1750 | | 8 | 8 | 15 | 10.3 | 6 | 6 | 6 | 6.0 |
| 6 | 1751 | Treatment F | 15 | 8 | 6 | 9.7 | 4 | 4 | 2 | 3.3 |
| | 1752 | | 15 | 15 | 15 | 15.0 | 8 | 8 | 6 | 7.3 |
| | 1753 | | 6 | 6 | 15 | 9.0 | 6 | 4 | 6 | 5.3 |
| | 1754 | | 8 | 8 | 15 | 10.3 | 6 | 4 | 6 | 5.3 |
| | 1755 | | 8 | 8 | 8 | 8.0 | 4 | 4 | 4 | 4.0 |
| | 1756 | | 8 | 6 | 6 | 6.7 | 4 | 4 | 4 | 4.0 |
| | 1757 | | 15 | 26 | 26 | 22.3 | 4 | 2 | 4 | 3.3 |
| | 1758 | | 8 | 6 | 6 | 6.7 | 4 | 6 | 4 | 4.7 |
| | 1759 | | 15 | 15 | 8 | 12.7 | 8 | 6 | 6 | 6.7 |
| | 1760 | | 8 | 15 | 15 | 12.7 | 6 | 4 | 4 | 4.7 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 6***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 8 | 8 | 6 | 7.3 | 8 | 6 | 4 | 6.0 |
| | 1762 | | 6 | 6 | 4 | 5.3 | 6 | 4 | 4 | 4.7 |
| | 1763 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 6 | 4.7 |
| | 1765 | | 15 | 15 | 8 | 12.7 | 8 | 4 | 4 | 5.3 |
| | 1766 | | 15 | 8 | 6 | 9.7 | 8 | 4 | 2 | 4.7 |
| | 1767 | | 8 | 15 | 8 | 10.3 | 4 | 2 | 2 | 2.7 |
| | 1768 | | 15 | 8 | 15 | 12.7 | 2 | 4 | 2 | 2.7 |
| | 1769 | | 8 | 8 | 15 | 10.3 | 4 | 4 | 2 | 3.3 |
| | 1770 | | 15 | 8 | 15 | 12.7 | 6 | 6 | 2 | 4.7 |
| | 1782 | | 8 | 15 | 8 | 10.3 | 15 | 8 | 4 | 9.0 |
| 8 | 1771 | Morphine sulfate | 6 | 6 | 4 | 5.3 | 6 | 4 | 4 | 4.7 |
| | 1772 | | 4 | 15 | 6 | 8.3 | 4 | 4 | 2 | 3.3 |
| | 1773 | | 8 | 6 | 6 | 6.7 | 6 | 4 | 2 | 4.0 |
| | 1774 | | 6 | 6 | 4 | 5.3 | 6 | 6 | 4 | 5.3 |
| | 1775 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1776 | | 15 | 15 | 15 | 15.0 | 2 | 2 | 2 | 2.0 |
| | 1777 | | 8 | 15 | 6 | 9.7 | 2 | 1.4 | 4 | 2.5 |
| | 1778 | | 15 | 8 | 15 | 12.7 | 6 | 2 | 4 | 4.0 |
| | 1779 | | 15 | 8 | 6 | 9.7 | 2 | 2 | 4 | 2.7 |
| | 1780 | | 8 | 6 | 6 | 6.7 | 4 | 2 | 2 | 2.7 |

## *Individual Animal Data – Mean Withdrawal Response (g) – Day 9*

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1701 | Treatment A | 6 | 6 | 6 | 6.0 | 4 | 2 | 4 | 3.3 |
| | 1702 | | 15 | 6 | 6 | 9.0 | 4 | 4 | 6 | 4.7 |
| | 1703 | | 8 | 8 | 8 | 8.0 | 2 | 2 | 2 | 2.0 |
| | 1704 | | 15 | 15 | 6 | 12.0 | 4 | 4 | 4 | 4.0 |
| | 1705 | | 15 | 15 | 8 | 12.7 | 8 | 4 | 4 | 5.3 |
| | 1706 | | 8 | 15 | 15 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1707 | | 8 | 8 | 15 | 10.3 | 6 | 2 | 4 | 4.0 |
| | 1708 | | 15 | 15 | 8 | 12.7 | 4 | 4 | 4 | 4.0 |
| | 1709 | | 8 | 8 | 15 | 10.3 | 4 | 6 | 4 | 4.7 |
| | 1710 | | 26 | 15 | 8 | 16.3 | 4 | 4 | 4 | 4.0 |
| 2 | 1711 | Treatment B | 15 | 6 | 8 | 9.7 | 15 | 8 | 6 | 9.7 |
| | 1712 | | 8 | 8 | 15 | 10.3 | 15 | 8 | 8 | 10.3 |
| | 1713 | | 15 | 6 | 8 | 9.7 | 8 | 8 | 8 | 8.0 |
| | 1714 | | 15 | 8 | 8 | 10.3 | 15 | 8 | 8 | 10.3 |
| | 1715 | | 8 | 8 | 8 | 8.0 | 8 | 26 | 8 | 14.0 |
| | 1716 | | 8 | 15 | 8 | 10.3 | 15 | 26 | 15 | 18.7 |
| | 1717 | | 6 | 15 | 8 | 9.7 | 8 | 15 | 8 | 10.3 |
| | 1718 | | 8 | 15 | 15 | 12.7 | 6 | 6 | 8 | 6.7 |
| | 1719 | | 8 | 8 | 6 | 7.3 | 15 | 8 | 8 | 10.3 |
| | 1720 | | 8 | 8 | 15 | 10.3 | 6 | 8 | 8 | 7.3 |
| 3 | 1721 | Treatment C | 8 | 6 | 8 | 7.3 | 4 | 2 | 2 | 2.7 |
| | 1722 | | 15 | 8 | 15 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1723 | | 6 | 6 | 4 | 5.3 | 15 | 8 | 6 | 9.7 |
| | 1781 | | 15 | 15 | 8 | 12.7 | 2 | 2 | 4 | 2.7 |
| | 1725 | | 15 | 15 | 6 | 12.0 | 6 | 2 | 4 | 4.0 |
| | 1726 | | 6 | 15 | 8 | 9.7 | 4 | 4 | 4 | 4.0 |
| | 1727 | | 4 | 8 | 15 | 9.0 | 4 | 2 | 4 | 3.3 |
| | 1728 | | 6 | 6 | 8 | 6.7 | 6 | 6 | 6 | 6.0 |
| | 1729 | | 8 | 8 | 8 | 8.0 | 4 | 4 | 6 | 4.7 |
| | 1730 | | 15 | 8 | 8 | 10.3 | 4 | 4 | 4 | 4.0 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 9***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1731 | Treatment D | 15 | 8 | 8 | 10.3 | 8 | 6 | 4 | 6.0 |
| | 1732 | | 15 | 8 | 8 | 10.3 | 4 | 2 | 2 | 2.7 |
| | 1733 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 2 | 3.3 |
| | 1734 | | 8 | 8 | 6 | 7.3 | 6 | 2 | 1.4 | 3.1 |
| | 1735 | | 8 | 6 | 6 | 6.7 | 4 | 4 | 2 | 3.3 |
| | 1736 | | 8 | 6 | 8 | 7.3 | 6 | 4 | 6 | 5.3 |
| | 1737 | | 8 | 8 | 8 | 8.0 | 2 | 2 | 2 | 2.0 |
| | 1738 | | 15 | 8 | 8 | 10.3 | 2 | 2 | 4 | 2.7 |
| | 1739 | | 8 | 6 | 6 | 6.7 | 4 | 4 | 4 | 4.0 |
| | 1740 | | 8 | 6 | 8 | 7.3 | 4 | 4 | 2 | 3.3 |
| 5 | 1741 | Treatment E | 8 | 6 | 8 | 7.3 | 6 | 6 | 4 | 5.3 |
| | 1742 | | 26 | 8 | 4 | 12.7 | 6 | 6 | 2 | 4.7 |
| | 1743 | | 8 | 8 | 8 | 8.0 | 4 | 2 | 2 | 2.7 |
| | 1744 | | 15 | 8 | 6 | 9.7 | 4 | 6 | 6 | 5.3 |
| | 1745 | | 15 | 8 | 15 | 12.7 | 4 | 4 | 6 | 4.7 |
| | 1746 | | 8 | 6 | 8 | 7.3 | 6 | 4 | 2 | 4.0 |
| | 1747 | | 15 | 6 | 8 | 9.7 | 6 | 8 | 4 | 6.0 |
| | 1748 | | 8 | 8 | 6 | 7.3 | 4 | 4 | 4 | 4.0 |
| | 1749 | | 15 | 15 | 8 | 12.7 | 6 | 6 | 4 | 5.3 |
| | 1750 | | 8 | 6 | 8 | 7.3 | 4 | 4 | 4 | 4.0 |
| 6 | 1751 | Treatment F | 15 | 8 | 6 | 9.7 | 6 | 6 | 4 | 5.3 |
| | 1752 | | 15 | 15 | 6 | 12.0 | 6 | 6 | 8 | 6.7 |
| | 1753 | | 8 | 8 | 6 | 7.3 | 6 | 4 | 2 | 4.0 |
| | 1754 | | 8 | 6 | 6 | 6.7 | 4 | 2 | 2 | 2.7 |
| | 1755 | | 8 | 6 | 6 | 6.7 | 4 | 6 | 4 | 4.7 |
| | 1756 | | 15 | 8 | 15 | 12.7 | 2 | 4 | 4 | 3.3 |
| | 1757 | | 8 | 8 | 15 | 10.3 | 4 | 6 | 6 | 5.3 |
| | 1758 | | 15 | 15 | 8 | 12.7 | 6 | 2 | 4 | 4.0 |
| | 1759 | | 8 | 8 | 8 | 8.0 | 15 | 8 | 6 | 9.7 |
| | 1760 | | 15 | 8 | 15 | 12.7 | 4 | 6 | 6 | 5.3 |

***Individual Animal Data – Mean Withdrawal Response (g) – Day 9***

| Group Number | Ear Tag Number | Treatment | Left Paw Reading 1 | Left Paw Reading 2 | Left Paw Reading 3 | Left Paw Mean | Right Paw Reading 1 | Right Paw Reading 2 | Right Paw Reading 3 | Right Paw Mean |
|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 1761 | Treatment G | 15 | 8 | 15 | 12.7 | 6 | 4 | 2 | 4.0 |
| | 1762 | | 6 | 8 | 8 | 7.3 | 6 | 4 | 2 | 4.0 |
| | 1763 | | 8 | 6 | 6 | 6.7 | 4 | 4 | 4 | 4.0 |
| | 1765 | | 15 | 15 | 8 | 12.7 | 6 | 2 | 4 | 4.0 |
| | 1766 | | 8 | 8 | 8 | 8.0 | 2 | 4 | 4 | 3.3 |
| | 1767 | | 26 | 8 | 15 | 16.3 | 4 | 4 | 4 | 4.0 |
| | 1768 | | 8 | 15 | 15 | 12.7 | 4 | 4 | 2 | 3.3 |
| | 1769 | | 15 | 8 | 8 | 10.3 | 4 | 4 | 4 | 4.0 |
| | 1770 | | 15 | 8 | 6 | 9.7 | 4 | 4 | 2 | 3.3 |
| | 1782 | | 8 | 8 | 15 | 10.3 | 6 | 4 | 4 | 4.7 |
| 8 | 1771 | Morphine sulfate | 15 | 8 | 8 | 10.3 | 4 | 4 | 2 | 3.3 |
| | 1772 | | 8 | 6 | 8 | 7.3 | 4 | 2 | 2 | 2.7 |
| | 1773 | | 8 | 8 | 15 | 10.3 | 6 | 2 | 2 | 3.3 |
| | 1774 | | 8 | 6 | 8 | 7.3 | 1.4 | 2 | 1.4 | 1.6 |
| | 1775 | | 8 | 6 | 15 | 9.7 | 6 | 4 | 1.4 | 3.8 |
| | 1776 | | 15 | 15 | 15 | 15.0 | 4 | 2 | 2 | 2.7 |
| | 1777 | | 6 | 4 | 4 | 4.7 | 4 | 2 | 2 | 2.7 |
| | 1778 | | 15 | 15 | 6 | 12.0 | 6 | 4 | 4 | 4.7 |
| | 1779 | | 8 | 8 | 6 | 7.3 | 2 | 2 | 6 | 3.3 |
| | 1780 | | 8 | 6 | 8 | 7.3 | 1.4 | 1 | 1 | 1.1 |

## B. Appendix II – Plasma Analysis

**Mouse IgG Quantification ELISA**

Antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti mouse IgG (gamma-chain specific) UNLB antibody (Southern Biotech 1030-01, 1mg/mL) was diluted 1:1000 in carbonate buffer (100mM NaHCO3, 33.6 mM Na2CO3, pH 9.5). 384 well plates were coated with 15µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tweenn-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting mouse anti-LPA antibody in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1: 7290, 1: 21870, 1: 65610, 1: 196380 and 1: 590490 and 15µL of the standard/samples preparation were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 µL of 1: 10,000 blocking buffer diluted goat anti-mouse IgG HRP (Southern Biotech 1030-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15 µl/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H2SO4, 15µl/well. The OD was measured at 450 nm. Data was analyzed using Graphpad Prism software. The standard ODs were graphed using a four parameter equation and this was used to calculate the mouse IgG in the samples. Excel software was used to calculate the µg/ml of mouse IgG in each sample by correcting the values for the dilution factor.

# RESULTS

## Antibody Concentration in plasma samples

| Group | Animal ID | (μg/mL) | Group | Animal ID | (μg/mL) |
|---|---|---|---|---|---|
| A | 1701 | 90.17 | E | 1741 | 126.61 |
| | 1702 | 188.44 | | 1742 | 231.93 |
| | 1703 | 133.59 | | 1743 | 226.59 |
| | 1704 | 184.39 | | 1744 | 195.38 |
| | 1705 | 210.40 | | 1745 | 178.47 |
| | 1706 | 270.66 | | 1746 | 149.21 |
| | 1707 | 185.78 | | 1747 | 334.75 |
| | 1708 | 197.92 | | 1748 | 143.46 |
| | 1709 | 273.33 | | 1749 | 295.63 |
| | 1710 | 139.70 | | 1750 | 149.38 |
| | **Average** | **187.44** | | **Average** | **203.14** |
| C | 1721 | 728.08 | F | 1751 | 570.15 |
| | 1722 | 517.97 | | 1752 | 622.99 |
| | 1723 | 623.88 | | 1753 | 402.17 |
| | 1724 | 815.34 | | 1754 | 712.65 |
| | 1726 | 657.53 | | 1755 | 791.53 |
| | 1727 | 1334.07 | | 1756 | 714.33 |
| | 1728 | 585.88 | | 1757 | 582.06 |
| | 1729 | 475.46 | | 1758 | 569.53 |
| | 1730 | 558.89 | | 1759 | 445.57 |
| | 1781 | 530.28 | | 1760 | 603.02 |
| | **Average** | **682.74** | | **Average** | **601.40** |
| D | 1731 | 253.37 | G | 1761 | 340.01 |
| | 1732 | 161.12 | | 1762 | 141.17 |
| | 1733 | 275.52 | | 1763 | 212.74 |
| | 1734 | 261.24 | | 1764 | 262.38 |
| | 1735 | 86.54 | | 1766 | 255.16 |
| | 1736 | 295.45 | | 1767 | 230.43 |
| | 1737 | 283.63 | | 1768 | 181.89 |
| | 1738 | 297.61 | | 1769 | 290.27 |
| | 1739 | 293.52 | | 1770 | 192.76 |
| | 1740 | 328.07 | | 1782 | 191.77 |
| | **Average** | **253.61** | | **Average** | **229.86** |

# DATA LOCATION

**Electronic Files:** Working folders/Stephanie/2014/PK/macro 10-1-14 IgG quant Calvert samples.xls
**Notebook#** 278 pages 38-39

## C. Appendix II – Protocol, Protocol Amendments and Deviation

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# Study Protocol

**Title:** Efficacy of IV LT3114 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

**Calvert Study No.:** 1274RL35.001

**Testing Facility:** Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

**Study Sponsor:** Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

# I. Table of Contents

# *II. Introduction*

## A. Title

Efficacy of IV LT3114 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

## B. Objective

The purpose of the study is to evaluate the potential ability of a test article to produce an analgesic response to post-surgical pain in rats.

## C. Regulatory Compliance

This is a non-regulated study. However, the study will be run in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert.

## D. Calvert Study Number

1274RL35.001

## E. Testing Facility

Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

## F. Sponsor

Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

## G. Study Director

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## H. Study Monitor

Rosalia Matteo, Ph.D.
Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404
Phone: (858) 678-0800 ext 112
Fax : (858)678-0900
E-mail: RMatteo@lpath.com

## I. Key Study Dates

| | |
|---|---|
| Proposed Study Start Date: | Feb 2014 |
| Proposed Experimental Start Date: | Mar 2014 |
| Proposed Experimental Completion Date: | Mar 2014 |

# *III. Materials and Methods*

## A. Test Articles, Reference Article and Vehicle

### *1. Test Article 1*

| | |
|---|---|
| Identification: | 1023 |
| Lot/Batch No.: | To be documented in the raw data and presented in the final report |
| Physical Description: | Clear, colorless solution |
| Storage Conditions: | Store stock vials at -70°C. Prepare fresh dilutions in PBS as needed prior to dosing, storing the remainder at 2-8°C for up to 1 week. Do not freeze/thaw. |

### 2. *Test Article 2*

Identification: 3114

Lot/Batch No.: To be documented in the raw data and presented in the final report

Physical Description: Clear, colorless solution

Storage Conditions: Store stock vials at -70°C. Prepare fresh dilutions in PBS as needed prior to dosing, storing the remainder at 2-8°C for up to 1 week. Do not freeze/thaw.

### 3. *Reference Article*

Identification: Morphine sulfate

Source/Manufacturer: To be documented in the raw data

Lot/Batch No.: To be documented in the raw data

Physical Description: Clear colorless liquid

Storage Conditions: Room temperature

### 4. *Vehicle*

Identification: Phosphate buffered saline (PBS)

Source/Manufacturer: To be documented in the raw data

Lot/Batch No.: To be documented in the raw data

Physical Description: Clear, colorless solution

Storage Conditions: Room temperature

### 5. *Dose Preparation*

The test article formulations will be provided as liquids by the Sponsor and dosed as received. The formulations will be stored at -70°C until use, after which working stocks will be stored at 2-8°C for up to 1 week.

## B. Test System (Animals and Animal Care)

### *1. Description*

| | |
|---|---|
| Species: | Rat |
| Strain/Substrain: | Hsd: SD or according to supplier |
| Total Number: | Eighty (80): ten (10) per group |
| Gender: | Male |
| Age Range: | 6 to 8 weeks |
| Body Weight Range: | 200-250 grams |
| Animal Source: | Harlan or any other approved supplier |
| Experimental History: | Purpose-bred and experimentally naïve at the outset of the study. |
| Identification: | Ear tag number and color code markings |

### *2. Rationale for Choice of Species and Number of Animals*

The rat is a standard species used for the evaluation of potential analgesic properties of a test article (1-6).

Based on Calvert's experience with this model and available literature, the number of animals used in this study is the minimum amount considered necessary to provide valid scientific assessment (4,5,6).

### *3. Husbandry*

Housing: Animals will be group housed upon receipt in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals will be kept will be documented in the study records. No other species will be kept in the same room. The experiment will be conducted in a room dedicated to this study and the dosing will occur

| | |
|---|---|
| | in this room only. Following surgery, animals will be individually housed in totes with bedding. |
| Lighting: | 12 hours light/12 hours dark. |
| Room Temperature: | 20 to 26°C |
| Relative Humidity: | 30-70% |
| Food: | All animals will have access to Certified Rodent Diet (TEKLAD) or equivalent *ad libitum*, unless otherwise specified. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which will be maintained in the Calvert archives. |
| Water: | Water will be available *ad libitum,* to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert's SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis will be maintained in the Calvert archives. |
| Acclimation: | Study animals will be acclimated to their housing for a minimum of 5 days prior to their day of dosing. |

## 4. *Pre-study Health Screen and Selection Criteria*

All animals received for this study will be assessed as to their general health by a member of the veterinary staff or other suitably trained authorized personnel. During the acclimation period, each animal will be observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

### 5. *Assignment to Study Groups*

Animals will be selected by body weight and apparent good health and will be randomly assigned to study groups.

### 6. *Humane Care of Animals*

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert's SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 2011, National Academy Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge this study is not unnecessary or unnecessarily duplicative.

## C. Test Article Administration

### 1. *Group Assignments and Dose Levels*

| Group | Surgery | Test Article | Dose (mg/kg in 1 ml/kg) | Route of Administration | Dose Schedule | Number of rats |
|---|---|---|---|---|---|---|
| 1 | Sham | PBS | 0 | IV | +3h, d3, d6 | 10 |
| 2 | Brennan | 1023 | 10 | IV | -1h, d3, d6 | 10 |
| 3 | Brennan | 3114 | 10 | IV | -1h, d3, d6 | 10 |
| 4 | Brennan | 1023 | 10 | IV | +3h, d3, d6 | 10 |
| 5 | Brennan | 3114 | 10 | IV | +3h, d3, d6 | 10 |
| 6 | Brennan | 1023 | 30 | IV | +3h, d3, d6 | 10 |
| 7 | Brennan | 3114 | 30 | IV | +3h, d3, d6 | 10 |
| 8 | Brennan | Morphine | 3 | SC | +3h, d3, d6 | 10 |

### 2. *Dosing*

| | |
|---|---|
| Route: | Intravenous (i.v.) injection for the vehicle and test articles<br><br>Subcutaneous (s.c.) injection for the reference article |
| Frequency: | Three times. 1 hour prior to or 3 hours after surgery, then twice a week (on Days 3 and 6) after behavioral testing. |
| Procedure: | Vehicle or test articles will be administered by intravenous injection at 1 mL/kg. Reference article will be administered by subcutaneous injection at 1 mL/kg. |

### 3. *Justification for Route and Dose Levels*

The route of administration was chosen as optimal based on pharmacokinetic studies of blood levels and half-life.

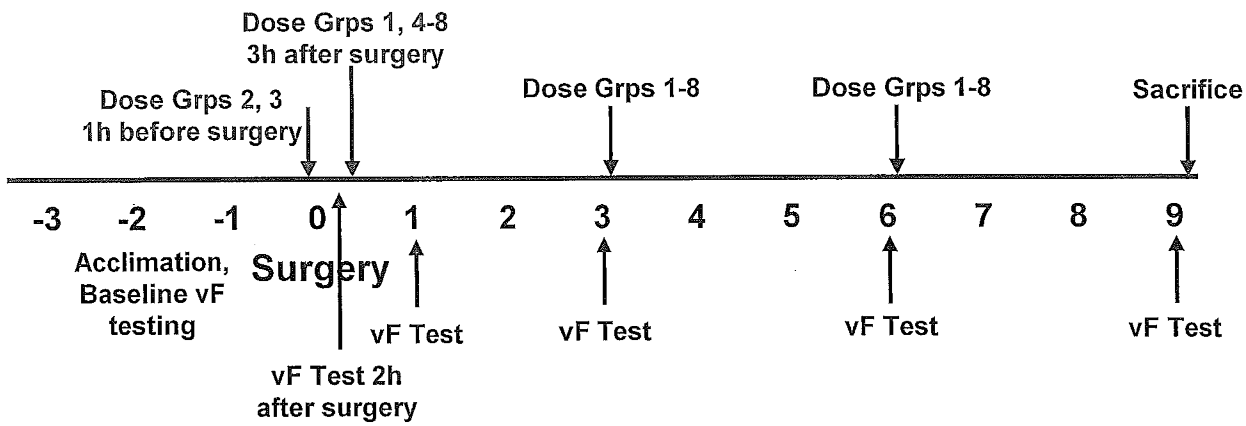
Doses were selected based on pharmacological efficacy studies in a variety of rat models of disease.

## D. Method of Study Performance

### 1. *Study Details:*

| Procedure | Time Points | Notes |
|---|---|---|
| Acclimation | 3-5 days before surgery | |
| von Frey behavioral testing | 1-3 days before surgery, then 2h, and on days 1, 3, 6, 9 after surgery | Also 30 min-2h after dosing morphine (see description) |
| Body weights | Prior to surgery, then every 2-3 days thereafter | |
| Antibody dosing | 1h prior to or 3h after surgery, then twice/week (Days 3 and 6) after behavioral testing | |
| Brennan or sham surgery | Day 0 | |
| Clinical observations | Daily for morbidity/mortality | |
| Report | Standard in-life summary | |
| Terminal Procedures | End of study | Collect blood for LPA measurement |
| Archive | | 2 years |

**Timeline (days):**

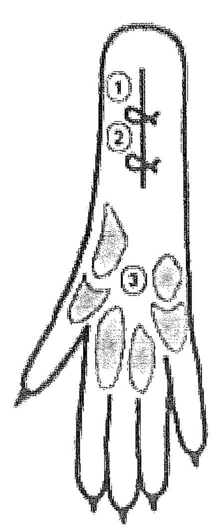



## 2. *Behavioral Testing:*

The rats will be placed individually on an elevated wire mesh floor covered with a clear plastic cage top and allowed to acclimate for ≥30 min to the von Frey testing apparatus and baseline tactile thresholds will be measured. Measurements of tactile paw withdrawal thresholds (PWT) will be conducted on both hindpaws by an observer blinded to the treatment conditions. Paw withdrawal responses will be determined using von Frey filaments applied from underneath the cage through openings. Each von Frey filament will be applied at the medial side of the wound near the heel (site 1, see diagram below) once starting with a 1 or 2 g filament and continuing until a withdrawal response occurred or 300 g force is reached. The median force producing a response, determined from three tests given over a 10-min period, will be considered the withdrawal threshold. Animals will be distributed evenly into groups according to their average baseline von Frey thresholds. Testing for morphine will occur 30, 60, 90 and 120 minutes after injection.

## 3. *Surgical Narrative*

Animals will be weighed prior to surgery and every 2-3 days thereafter. Surgical procedures will be performed under deep isoflurane anesthesia (3-4% at 3-4 L/min $O_2$). All surgical procedures will be performed at the surgical planes of anesthesia, using sterile instruments that are re-sterilized for each animal. After disinfection of the right hind paw with Betadine (povidone-iodine) and 70% isopropyl alcohol, rats will receive an intramuscular injection of penicillin (30,000 IU) in the triceps muscle. A 1 cm longitudinal incision will be made with a



number 11 blade through the skin and fascia of the right hind paw starting 0.5 cm from the proximal edge of the heel and extending toward the toes (1). Following skin incision, blunt curved forceps will be used to raise and retract the plantar flexor digitorum brevis muscle. The plantaris muscle is then incised longitudinally, keeping the muscle origin and insertion intact. The skin will be closed with 2 mattress sutures of 5-0 silk, and the wound site will be covered with triple antibiotic ointment (polymixin B, neomycin and bacitracin). To minimize distress, fluids (s.c. Plasmalyte – 1 ml/50g) will be administered immediately post-operatively. Sham surgeries will consist of all procedures except incision and muscle retraction (anesthesia, antibiotics and surgical preparation). After surgery, the animals will be allowed to recover in a box with a heating pad so as to prevent rapid loss of body heat following surgery. On the second postoperative day, sutures will be removed according to Brennan et al. Animals will be monitored daily for general health. Any animals showing evidence of motor dysfunction, > 20% weight loss or wound dehiscence after surgery will be euthanized.

### *4. Blood Collection*

Immediately following the behavioral testing on Day 9, the rats will be anesthetized by $CO_2$ inhalation and whole blood samples (approximately 0.5 ml/sample) will be collected by cardiocentesis into blood collection tubes containing EDTA. Immediately following blood collection, the animals will euthanized by $CO_2$ asphyxiation. Blood samples will be spun in a centrifuge (3000 rpm for approximately 10 minutes). After centrifugation, plasma will be collected and stored at -80°C ± 10°C until shipped for analysis. The Sponsor will provide contact information for shipping the samples.

## E. Terminal Procedures

### *1. Termination*

a) Scheduled Sacrifice

All surviving animals will be euthanized by $CO_2$ asphyxiation immediately following the blood collection on Day 9.

b) Unscheduled Sacrifice

Animals may be euthanized by $CO_2$ asphyxiation for humane reasons based upon criteria outlined in Calvert's SOP VET-14, "Criteria for Determination of Moribund Animals and for Sacrifice for Humane Reasons"

## *IV. Records and Reports*

### A. Statistical Analysis

Mean and standard error of the mean will be calculated for paw withdrawal thresholds. Statistically significant effects will be determined using an ANOVA with Tukey HSD Multiple Comparison Test (Systat V. 9.01). A p value $\leq 0.05$ will denote statistical significance.

### B. Storage of Records

In-life data, protocol, protocol amendments (if applicable), and the final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

Six months following the submission of the audited draft report if there are no comments generated by the Sponsor and/or Study Monitor, the Sponsor/Study Monitor will be notified and the report will be finalized and archived according to the terms stated in the protocol.

## *V. Miscellaneous*

### A. References

1. Brennan, T.J., Vandermeulen, E.P. and Gebhart, G.F. Characterization of a rat model of incisional pain. Pain **64**: 493-501 (1996).

2. Hamalainen, M.M., Gebhart, G.F. and Brennan, T.J. Acute effects of an incision on mechanosensitive afferents in the plantar rat hindpaw. *J Neurophysiol* **87**: 712-720(2002).

3. Zahn, P.K. and Brennan, T.J. Primary and secondary hyperalgesia in a rat model for human post-operative pain. Anesthesiology **90**: 863-872(1999).

4. Whiteside, G.T., Harrison, J., Boulet, J., Mark, L., Pearson, M., Gottshall, S. and Walker, K., Pharmacological characterization of a rat model of incisional pain. British Journal of Pharmacology **141**: 85-91(2004).

5. Xu, J. and Brennan, T.J., Guarding Pain and Spontaneous Activity of Nociceptors after Skin versus Skin Plus Deep Tissue Incision, Anesthesiology 112:153-164 (2010).

6. Brennan, T.J., Postoperative Models of Nociception, ILAR Journal **40(3)**:129-136 (1999).

## VI. Protocol Approval Signatures

Brian Klatt
21 Feb 2014

Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

Date

21 Feb 2014

Scientific Management
Calvert Laboratories, Inc.

Date

20 FEB 2014

IACUC
Calvert Laboratories, Inc.

Date

Feb. 19. 2014

Rosalia Matteo, Ph.D.
Study Monitor
Lpath, Inc.

Date

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PROTOCOL AMENDMENT

CALVERT STUDY NO.: 1274RL35.001

STUDY TITLE: Efficacy of IV 504B3 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

AMENDMENT NUMBER: 1

EFFECTIVE DATE: 2 May 2014

PROTOCOL SECTION CHANGE (pages 1 and 4):

**Title:** Efficacy of IV ~~LT3114~~ **504B3** on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

PROTOCOL SECTION CHANGE (page 5):

***III. Materials and Methods***

**A. Test Articles, Reference Article and Vehicle**

**1. Test Article 1**

Identification: ~~1023~~ **LT1015**

**2. Test Article 2**

Identification: ~~3114~~ **504B3**

PROTOCOL SECTION CHANGE (page 9):

## C. Test Article Administration

### *1. Group Assignments and Dose Levels*

| Group | Surgery | Test Article | Dose (mg/kg in 1 ml/kg) | Route of Administration | Dose Schedule | Number of rats |
|---|---|---|---|---|---|---|
| 1 | Sham | PBS | 0 | IV | +3h, d3, d6 | 10 |
| 2 | Brennan | ~~1023~~ **LT1015** | 10 | IV | -1h, d3, d6 | 10 |
| 3 | Brennan | ~~3114~~ **504B3** | 10 | IV | -1h, d3, d6 | 10 |
| 4 | Brennan | ~~1023~~ **LT1015** | 10 | IV | +3h, d3, d6 | 10 |
| 5 | Brennan | ~~3114~~ **504B3** | 10 | IV | +3h, d3, d6 | 10 |
| 6 | Brennan | ~~1023~~ **LT1015** | 30 | IV | +3h, d3, d6 | 10 |
| 7 | Brennan | ~~3114~~ **504B3** | 30 | IV | +3h, d3, d6 | 10 |
| 8 | Brennan | Morphine | 3 | SC | +3h, d3, d6 | 10 |

REASON FOR CHANGE:

Due to drug availability it was decided to use the murine version of the antibodies. 1023 and 3114 are the human antibodies. LT1015 will substitute 1023 and 504B3 to 3114.

SIGNATURES:

Brian Klatt

Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

5 May 2014
Date

Steven D. Stoneham

Scientific Management
Calvert Laboratories, Inc.

5 May 2014
Date

William G. Tieman

IACUC
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5 May 2014
Date

R Matteo

Rosalia Matteo, Ph.D.
Study Monitor
Lpath, Inc.

05-02-2014
Date

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PROTOCOL AMENDMENT

CALVERT STUDY NO.: 1274RL35.001

STUDY TITLE: Efficacy of IV 504B3 on Primary Hyperalgesia in the Brennan Model of Post-Incisional Pain in Rats (non-GLP)

AMENDMENT NUMBER: 2

EFFECTIVE DATE: 8 May 2014

PROTOCOL SECTION CHANGE (page 6):

## *III. Materials and Methods*

### A. Test Articles, Reference Article and Vehicle

#### 5. Dose Preparation

The test article formulations will be provided as liquids by the Sponsor and dosed as received **for the 30 mg/kg dose groups. For the 10 mg/kg dose groups, the stocks solutions will be diluted to concentrations that would allow the same dose volumes as the 30 mg/kg dose groups**. The formulations will be stored at -70°C until use, after which working stocks will be stored at 2-8°C for up to 1 week.

REASON FOR CHANGE:

To clarify the dose preparation based on the supplied stock solution concentrations and the desire to keep the dose volumes the same for the 10 mg/kg groups as the 30 mg/kg groups.

PROTOCOL SECTION CHANGE (page 9):

## *III. Materials and Methods*

### C. Test Article Administration

### *1. Group Assignments and Dose Levels*

| Group | Surgery | Test Article | Dose (mg/kg ~~in 1 ml/kg~~) | **Dose Volume (mL/kg)** | Route of Administration | Dose Schedule | Number of rats |
|---|---|---|---|---|---|---|---|
| 1 | Sham | PBS | 0 | **2.40** | IV | +3h, d3, d6 | 10 |
| 2 | Brennan | LT1015 | 10 | **2.35** | IV | -1h, d3, d6 | 10 |
| 3 | Brennan | 504B3 | 10 | **2.40** | IV | -1h, d3, d6 | 10 |
| 4 | Brennan | LT1015 | 10 | **2.35** | IV | +3h, d3, d6 | 10 |
| 5 | Brennan | 504B3 | 10 | **2.40** | IV | +3h, d3, d6 | 10 |
| 6 | Brennan | LT1015 | 30 | **2.35** | IV | +3h, d3, d6 | 10 |
| 7 | Brennan | 504B3 | 30 | **2.40** | IV | +3h, d3, d6 | 10 |
| 8 | Brennan | Morphine | 3 | **1.0** | SC | +3h, d3, d6 | 10 |

### *2. Dosing*

Route: Intravenous (i.v.) injection for the vehicle and test articles. **The injection will be administered as a slow bolus over approximately 30 seconds.**

Subcutaneous (s.c.) injection for the reference article

Procedure: Vehicle or test articles will be administered by intravenous injection at ~~1~~ **2.35 or 2.40** mL/kg. Reference article will be administered by subcutaneous injection at 1 mL/kg.

REASON FOR CHANGE:

The dose volume column was added to reflect the change in dosing volumes to accommodate the desired doses using the supplied stock concentrations of LT1015 (12.74 mg/mL) and 504B3 (12.48 mg/mL). Since the dose volumes were increased it was noted that the injection would be administered as a slow bolus.

SIGNATURES:

Brian Klatt — 8 May 2014

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Study Director
Calvert Laboratories, Inc.

Date

8 May 2014

Scientific Management
Calvert Laboratories, Inc.

Date

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05-08-2014

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## Protocol Deviation

1. The actual body weight range (192-229 grams) deviated from the protocol-stated of 200-250 grams. Only two rats had body weights less than 200 grams. Rat 1762 weighed 192 grams and rat 1773 weighed 195 grams. The protocol deviation had no impact on the outcome of the study since the deviation was of a small magnitude and since the animals were dosed according to body weight.